Improved EMCF-SBAS Processing Chain Based on Advanced Techniques for the Noise-Filtering and Selection of Small Baseline Multi-Look DInSAR Interferograms

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***Abstract*—We present in this paper a solution to drastically improve the deformation time-series retrieval capability of the small baseline differential SAR interferometry (DInSAR) processing chain based on the cascade of the extended minimum cost flow (EMCF) phase unwrapping method and of the small baseline subset (SBAS) inversion technique. This improvement relies on the inclusion of two preprocessing steps implementing an effective noise-filtering operation and an efficient interferogram selection procedure, respectively. The former step filters out the noise affecting the phase components of a redundant set of conventional multi-look small baseline interferograms. This is achieved by solving, for each pixel, a nonlinear minimization problem based on computing the wrapped phase vector that minimizes the weighted circular variance of the phase difference between the original and noise-filtered interferograms. This technique is very easy to implement because it does not require any pixel selection step to be applied to the exploited full-resolution SAR images, and it has no need of any *a priori* information on the statistics of the complex-valued SAR images. The latter step, implementing the interferogram selection procedure, is carried out via a computationally efficient simulated annealing algorithm and allows identifying the optimum set of previously filtered small baseline interferograms to be used as input for the original EMCF-SBAS processing chain by maximizing the (average) coherence values. The presented results, achieved by processing three data sets collected by the ENVISAT ASAR sensor over the Abruzzi region (Central Italy), Mt. Etna volcano (South Italy), and Yellowstone Caldera (WY, USA), demonstrate the effectiveness of the proposed advanced EMCF-SBAS processing chain.**

***Index Terms*—Deformation time-series, differential synthetic aperture radar interferometry (DInSAR), small baseline subset (SBAS).**

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I. Introduction

DIFFERENTIAL synthetic aperture radar interferometry (DInSAR) has evolved over the years to become, nowadays, a well-established remote sensing methodology for Earth surface deformation detection and monitoring [1], [2]. The DInSAR technique exploits the phase difference (interferogram) between two SAR acquisitions collected at different epochs and provides a measurement of the ground displacement projection along the radar line of sight (LOS), with a centimeter-to-millimeter accuracy [3]. While originally applied to investigate single deformation episodes [4]–[8], the DInSAR methodology has progressively (and successfully) been extended to detect and investigate the temporal evolution of the retrieved surface displacements via the development of "advanced" DInSAR techniques [9]–[15]. These algorithms are based on the inversion of properly generated sequences of DInSAR interferograms and allow the computation of LOS-projected deformation time-series in areas that are not significantly affected by decorrelation noise [16], [17].

The advanced DInSAR techniques are usually grouped into two main categories: the persistent scatterer (PS) methods [9]–[12], which are mostly focused on investigating point targets, and the small baseline (SB) techniques [13]–[15] that also analyze distributed targets (DS). In particular, the PS methods operate on the pixels of the full-resolution (single-look) images that are representative of resolution cells characterized by a dominant scatterer. These targets are not significantly affected by spatial and temporal decorrelation [16], thus permitting us to analyze interferometric SAR image pairs also with a very large spatial and temporal separation (baseline) between the acquisition orbits. Accordingly, the interferometric pairs can be chosen without imposing any constraint on their baselines, and all of the interferograms can be generated with respect to one single common SAR scene, usually referred to as "master image."

Conversely, the SB techniques also allow the exploitation of DSs that may be affected by spatial and temporal decorrelation phenomena. Accordingly, a proper selection of interferometric SAR data pairs with limited spatial and temporal baselines is carried out. As a consequence, the selection of a single master image for all of the interferograms of a data set is unfeasible, and a multiple master strategy has to be assumed [13]–[15].

Among the SB techniques, a well-known approach is the one referred to as small baseline subset (SBAS) [13] that allows

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producing LOS-projected mean deformation velocity maps and corresponding displacement time-series. The SBAS technique has originally been developed to analyze multi-look (complex average) interferograms [13], and it has subsequently been extended to work with full-resolution interferograms [18], with the aim to also detect localized deformation signals that may affect, for instance, single buildings or man-made structures. As previously said, the SBAS approach relies on the use of multiple master interferograms generated by imposing constraints on the maximum allowed spatial and temporal baselines. Accordingly, the baseline selection implies that the SAR data pairs used to produce the interferograms may be arranged in (few) subsets that are separated by large baselines. These subsets turn out to be independent from each other (since no suitable interferogram that connects SAR images belonging to different subsets exists), thus giving rise to an underdetermined problem for the time-series inversion, which can be solved by searching for a minimum norm least squares solution via the singular value decomposition (SVD) method [13].

We further remark that the SBAS technique is fully compatible with the phase unwrapping (PhU) procedure referred to as extended minimum cost flow (EMCF) [19]. This technique represents the space–time extension of the original minimum cost flow (MCF) PhU approach [20] and is based on the joint exploitation of the spatial and temporal relationships among the computed interferograms. The EMCF approach involves the computation of two Delaunay triangulations, usually referred to as "temporal" and "spatial" triangulations, respectively. The former takes into account the SAR data acquisition distribution in the temporal/perpendicular baseline plane [19], [21]. From the original temporal Delaunay triangulation, all of the triangles containing at least one "large baseline" interferogram are discarded. This leads to a reduced-Delaunay triangulation, which identifies the time-redundant network of small baseline DInSAR interferograms to be subsequently exploited. The spatial triangulation, instead, involves a selection, in the azimuth/range plane, of the coherent pixels that are common to the selected small baseline interferograms. It is worth remarking that, owing to the existence of high-performance routines implementing the MCF problem solver [22], the EMCF approach is computationally efficient with respect to other recently proposed space/time PhU approaches [23]–[25] that, instead, require the solution of time-consuming linear programming problems.

The joint exploitation of the EMCF and SBAS procedures, synthetically referred hereinafter to as EMCF-SBAS processing chain, has demonstrated its effectiveness to detect and investigate the temporal evolution of surface deformation phenomena in different applications. For instance, it has widely been exploited to analyze SAR data acquired by the C-band (5.6-cm wavelength) sensors onboard the ERS-1/2 and the ENVISAT platforms [26]–[30] of the European Space Agency (ESA). Note also that, when these two radar systems (operating with slightly different carrier frequencies) are collecting data with the same electromagnetic characteristics and illumination geometries, the ERS and ENVISAT data sets may be also jointly exploited by assuming that they belong to independent subsets that can be easily "linked" with one another by applying the original SBAS strategy [31].

The EMCF-SBAS processing chain has also been used to generate displacement time-series from sequences of SAR images of the C-band Canadian Space Agency RADARSAT-1 sensor [32]. More recently, it has been applied to the X-band (3.1-cm wavelength) data acquired by the new generation SAR instruments onboard the COSMO-SkyMed [33]–[35] sensor constellation of the Italian Space Agency (ASI), which combine improved spatial resolutions and shorter revisit times with respect to those permitted by the previous generation C-band radar sensors.

We present in this paper an advanced version of the EMCF-SBAS processing chain, which drastically improves the deformation time-series retrieval capability of the original EMCF-SBAS procedure. This improvement is achieved by complementing the original processing chain with two additional preprocessing steps, with the former implementing an effective interferogram noise-filtering operation and the latter an efficient interferogram selection, which allows us to identify the optimal reduced temporal triangulation to be exploited by the EMCF-SBAS processing chain.

The effectiveness of the proposed advanced EMCF-SBAS processing chain is demonstrated through an extensive set of experiments carried out on three sequences of SAR images collected by the ASAR instrument onboard the ENVISAT satellite.

II. Rationale of the Advanced EMCF-SBAS Processing Chain

We summarize in this section the main processing elements characterizing the proposed advanced EMCF-SBAS processing chain. To this aim, we start by considering the original EMCF-SBAS processing chain, sketched in Fig. 1, consisting of the previously mentioned two steps implementing the EMCF and the SBAS procedures, respectively. As said, the input of the original EMCF-SBAS processing chain is represented by a sequence of small baseline multi-look interferograms identified by a reduced-Delaunay triangulation computed in the temporal/perpendicular baseline plane. These interferograms are first unwrapped through the EMCF technique and then inverted via the SBAS algorithm[1] to compute the mean deformation velocity maps and, above all, surface displacement time-series in correspondence to the temporally coherent pixels [19]. The advanced EMCF-SBAS processing chain, whose block diagram is shown in Fig. 2, complements the original EMCF-SBAS processing chain with two additional preprocessing steps referred to as interferogram noise-filtering and interferogram selection techniques, respectively.

The former, briefly presented first in [36] and extensively described in the following Section III, allows us to significantly mitigate the decorrelation noise effects in a sequence of time-redundant multi-look small baseline interferograms, identified by simply imposing constraints on the maximum allowed spatial and temporal baselines, by properly exploiting their temporal relationships. In particular, this result is achieved as follows: 1) by solving a nonlinear minimization problem involving only the phase information of the exploited interferograms and 2) without requiring the implementation of any pixel-by-pixel adaptive procedures for the DS identification, based on the distributed scattering hypothesis [38], [39] and

[1]The SBAS inversion also includes a decoupling of the required deformation signal component from the undesired phase patterns, referred to as topographic and atmospheric artifacts [13], [37].

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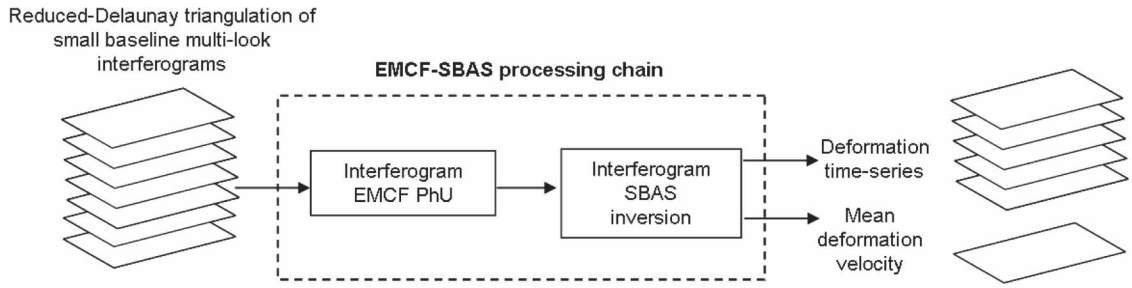


Fig. 1. Block diagram of the original EMCF-SBAS processing chain.

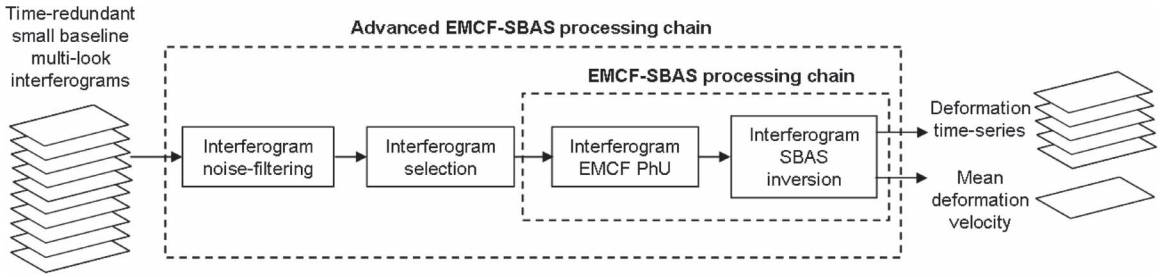


Fig. 2. Block diagram of the proposed advanced EMCF-SBAS processing chain.

applied to the full-resolution SAR images. Note that these are distinct and easy-to-implement features of our approach with respect to the phase linking approach [39], the SqueeSAR algorithm [40], and other recently proposed multitemporal interferogram filtering techniques [41]–[43]. Moreover, another peculiar characteristic of our method is the exploitation of a limited set of small baseline interferograms, which are selected by simply imposing constraints on the maximum allowed interferometric spatial and temporal baselines.

For what concerns the interferogram selection technique, which is extensively discussed in Section IV, it is aimed at identifying, starting from the previously filtered multi-look small baseline interferograms, the most appropriate temporal reduced triangulation to be exploited by the EMCF technique. In particular, our procedure maximizes the (average) coherence values of the small baseline multi-look interferograms belonging to the selected reduced triangulation through a simulated annealing (SA) searching strategy [44].

III. Interferogram Noise-Filtering Technique

We present in this section the analytical description of the proposed interferogram noise-filtering algorithm. Moreover, in order to show the effectiveness of the developed technique, we also present some results achieved by considering a data set of ENVISAT SAR images acquired over the Abruzzi region (Central Italy), which represents a very critical test-site since the related interferograms are typically affected by severe decorrelation noise artifacts.

A. Algorithm

Let us assume the availability of a set of $N+1$ SAR images acquired over the same area at ordered epochs $\mathbf{t} \equiv [t_0, t_1, \ldots, t_N]^T$, properly coregistered [45] with respect to the master scene collected at t_m, which is used as reference for the temporal $\mathbf{T} \equiv [t_0 - t_m, t_1 - t_m, \ldots, t_N - t_m]^T$ and spatial $\mathbf{b}_\perp \equiv [b_{\perp 0}, b_{\perp 1}, \ldots, b_{\perp N}]^T$ baseline vectors, with the latter relevant to the perpendicular component with respect to the radar LOS (perpendicular baseline). From these $N+1$ images, we first identify the M SAR data pairs characterized by small perpendicular and temporal baselines, and then, we generate the corresponding sequence of M small baseline multi-look interferograms, e.g., $\Delta\Phi(x,r) = [\Delta\Phi_1(x,r), \Delta\Phi_2(x,r), \ldots, \Delta\Phi_M(x,r)]^T$, with (x,r) being the azimuth and range coordinates of a generic radar pixel, and $\Delta\Phi_k(x,r) = \langle \Phi_{IM_k}(x,r) - \Phi_{IS_k}(x,r) \rangle_{-\pi,\pi} \; \forall\, k = 1, \ldots, M$, where $\Phi(x,r) = [\Phi_0(x,r), \Phi_1(x,r), \ldots, \Phi_N(x,r)]^T$ is the vector of the (unknown) wrapped phase values relevant to the $N+1$ SAR scenes. Note also that IM_k and IS_k, $\forall\, k = 1, \ldots M$, are the two indexes corresponding to the (master) and (slave) time acquisitions of the kth SAR data pair, respectively, and the symbol $\langle \cdot \rangle_{-\pi,\pi}$ stands for the wrapping operator. Note that the dependence on (x,r) variables is hereinafter not explicitly mentioned since the proposed technique implies a pixel-by-pixel temporal analysis.

The presented noise-filtering approach arises from the observation that multi-look interferograms are not fully time-consistent because they are typically generated through the multi-look operation, followed by a noise-filtering step [46], [47], and these steps are both independently carried out on each single interferogram. Indeed, the sum of the three (generic) multi-look interferograms computed from the SAR images acquired at the epochs t_A, t_B, and t_C, i.e., $\Delta\Phi_{BA} = \langle \Phi_B - \Phi_A \rangle_{-\pi,\pi}$, $\Delta\Phi_{CB} = \langle \Phi_C - \Phi_B \rangle_{-\pi,\pi}$, and $\Delta\Phi_{AC} = \langle \Phi_A - \Phi_C \rangle_{-\pi,\pi}$, is different from zero [40], [48]

$$\langle \Delta\Phi_{BA} + \Delta\Phi_{CB} + \Delta\Phi_{AC} \rangle_{-\pi,\pi} \neq 0. \tag{1}$$

Our noise-filtering method is based on analyzing the temporal inconsistencies of the computed time-redundant network of multi-look interferograms; it first allows estimating the (unknown) vector of the wrapped phases (related to the SAR acquisitions), which is subsequently used to reconstruct a new sequence of time-consistent noise-filtered interferograms. The

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exploitation of conventional small baseline multi-look interferograms is the distinctive characteristic of the proposed approach with respect to previous solutions, such as the SqueeSAR method [40] and other recently proposed multitemporal filtering techniques based on constraining the analysis to distributed scatterers [41], [42], which are identified through a pixel-by-pixel selection procedure performed at the full-resolution complex SAR image spatial grid. Such a selection permits us to rely on the distributed scattering hypothesis[2] under which the probability density function (pdf) of the complex-valued SAR image may be regarded as being a zero-mean multivariate circular normal distribution, and an appropriate maximum likelihood (ML) estimation step of the filtered phase values associated to each SAR acquisition is implemented [40].

On the contrary, in our approach, we focus on conventional multi-look interferograms generated without any *a priori* pixel selection step. Accordingly, we cannot rely on the validity of the aforementioned distributed scattering hypothesis. This is not a limiting issue for our technique that, for each pixel of the multi-look grid, is based on searching for the (unknown) phase vector of the wrapped phases (related to the SAR acquisitions), e.g., $\hat{\Phi}$, that minimizes the weighted circular variance [49]–[53], e.g., ν, of the random phase vector, namely, $\mathbf{r}$, representative of the phase difference between the original and reconstructed small baseline interferograms (residual phases). In particular, our solution leads to the formulation of the following nonlinear minimization problem:

$$\hat{\Phi} = \arg\min\left[\nu(\hat{\Phi})\right] = \arg\min\left[1 - \left|\Pi(\hat{\Phi})\right|\right] \tag{2a}$$

$$\Pi(\hat{\Phi}) = \sum_{k=1}^{M} w_k \exp[j \cdot r_k] \Big/ \sum_{k=1}^{M} w_k = \rho(\hat{\Phi}) \exp\left[j\mu(\hat{\Phi})\right] \tag{2b}$$

where $j = \sqrt{-1}$, $r_k = \langle \Delta\Phi_k - (\hat{\Phi}_{IM_k} - \hat{\Phi}_{IS_k}) \rangle_{-\pi,\pi}$, $\forall\, k = 1, \ldots, M$, are the residual phases and w_k, $\forall\, k = 1, \ldots, M$, are the weights representing our confidence on the phase quality of the exploited small baseline interferograms. Note also that ρ and μ represent the mean resultant length and the mean direction of the weighted resultant vector Π associated to the residual phases [49]–[53], respectively. Moreover, ρ can be also seen as a weighted version of the temporal coherence factor [19], which is, instead, computed as the difference between the original interferograms and the ones reconstructed from the deformation time-series achieved following the SBAS inversion. In addition, the weights w_k, $\forall\, k = 1, \ldots, M$, in (2b) are set by estimating the spatial coherences computed from the phase of the original multi-look interferograms $\boldsymbol{\Delta\Phi}$ as follows [36]:

$$w_k(x,r) = \frac{1}{(2N_A+1)(2N_R+1)} \times \left| \sum_{h=-N_A}^{N_A} \sum_{p=-N_R}^{N_R} \exp\left[j \cdot \Delta\Phi_k(x+h, r+p)\right] \right| \quad \forall\, k = 1, \ldots, M \tag{3}$$

where $2N_A + 1$ and $2N_R + 1$ are the number of azimuth and range pixels within the used boxcar averaging window, which is centered around the generic pixel of coordinates (x, r).

It is straightforward to demonstrate (see Appendix I) that the solution $\hat{\Phi}$ of the problem in (2a) and (2b) is the one that minimizes the weighted dispersion of the vector $\mathbf{r}$ with respect to the corresponding weighted mean direction (i.e., μ) [50]–[53].

Once the solution of the problem stated in (2a) and (2b) is retrieved, the $\hat{\Phi}$ phase values are then paired to reconstruct the corresponding sequence of noise-filtered interferograms $\Delta\hat{\Phi} = [\Delta\hat{\Phi}_1, \ldots, \Delta\hat{\Phi}_M]^T$, where $\Delta\hat{\Phi}_k = \langle \hat{\Phi}_{IM_k} - \hat{\Phi}_{IS_k} \rangle_{-\pi,\pi}$ $\forall\, k = 1, \ldots, M$. The experimental results shown in the following Section III-B make clear the very remarkable improvement of the spatial coherence values of the generated multi-look interferograms with respect to the original ones.

Nevertheless, a group of the reconstructed interferograms, although limited, can exhibit spatial coherence values smaller than the ones relevant to the corresponding original interferograms (see again Section III-B), thus implying that a partial corruption of the spatial coherence occurs during the minimization procedure. In particular, it happens in correspondence to interferograms that were originally significantly coherent, and this is due to the fact that the problem in (2a) and (2b) tends to "inject" coherence from very coherent to incoherent interferograms by exploiting the time redundancy of the small baseline data pair network.

To also preserve the spatial coherence of the very coherent interferograms and further increase the phase quality of the whole set of M reconstructed interferograms, we carry out a simple nonlinear combination between the original $\Delta\Phi$ and reconstructed $\Delta\hat{\Phi}$ interferograms. Accordingly, we combine the two sets of interferograms through the following (wrapped) weighted averaging operation:

$$\Delta\widehat{\Phi}_k = \arctan\left(\frac{w_k \sin(\Delta\Phi_k) + \hat{w}_k \sin(\Delta\hat{\Phi}_k)}{w_k \cos(\Delta\Phi_k) + \hat{w}_k \cos(\Delta\hat{\Phi}_k)}\right) \quad \forall k = 1, \ldots, M \tag{4}$$

where $\hat{w}_k(\cdot)$ is the spatial coherence computed from the phase of the kth reconstructed multi-look interferogram, simply estimated from (3) by replacing $\Delta\Phi_k$ with $\Delta\hat{\Phi}_k$. Following the (weighted) averaging operation in (4), the set of the so-computed M interferograms $\Delta\widehat{\Phi} \equiv [\Delta\widehat{\Phi}_1, \Delta\widehat{\Phi}_2, \ldots, \Delta\widehat{\Phi}_M]$ is characterized by increased levels of coherence (see Section III-B). At this stage, it is easy to recognize that this new sequence of multi-look interferograms $\Delta\widehat{\Phi}$ is, in turn, not fully time-consistent as the original one $\Delta\Phi$ was, although the interferogram temporal inconsistency has strongly been reduced. Accordingly, we may consider the possibility, at the expenses of the overall computation time of the algorithm, to carry out some successive iterations by applying the procedure in (2a) and (2b) to the sequences of time-inconsistent (average) noise-filtered differential interferograms computed following (4). The block diagram of the presented interferogram noise-filtering technique is shown in Fig. 3.

A few additional considerations on the proposed interferogram noise-filtering procedure are now in order.

First, we underline that our interferogram noise-filtering technique is not just an adaptation of the phase linking approach

[2]This condition is met when no dominant scatterers (such as artificial objects) are present in the resolution cell [38].

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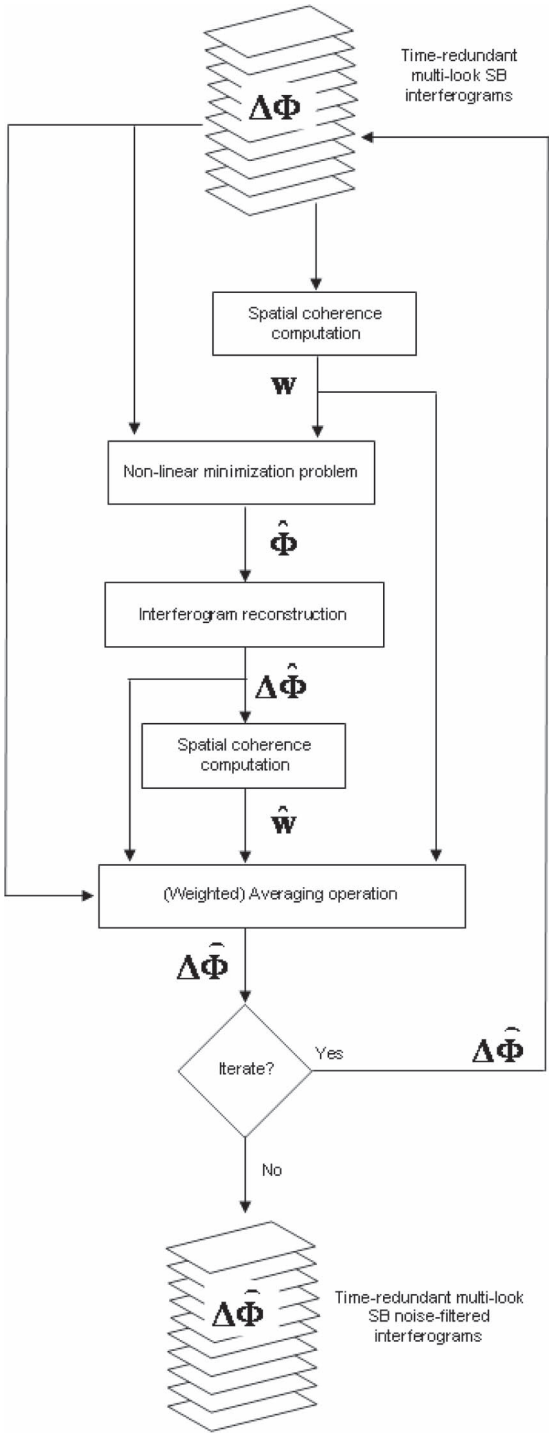


Fig. 3. Block diagram of the proposed interferogram noise-filtering technique.

and of the SqueeSAR algorithm [40] to the SBAS context. Indeed, although these methods have the same starting point, consisting in the lack of triangularity[3] among the phases due to the inconsistencies in the multi-look interferograms [see (1)], they are fundamentally different for what concerns both the exploited statistical frameworks and the mathematical formulation of the minimization procedures applied to retrieve the filtered interferograms. These issues are discussed in detail in Appendix II.

Second, we remark that the minimization of the nonlinear functional in (2a) and (2b) can be independently performed for each pixel of the multi-look grid by using one of the several nonlinear optimization software packages currently available in literature [54]–[56]. In particular, in this paper, we use the quasi-Newton limited-memory Broyden–Fletcher–Goldfarb–Shanno method [55] (freely available at http://users.eecs.northwestern.edu/~nocedal/lbfgsb.html), which is particularly suitable to solve constrained/unconstrained large nonlinear optimization problems, by guaranteeing stable results and high computational efficiency.

Third, it is worth noting that, since only phase differences appear in (2a) and (2b), the obtained solution can be only estimated up to a constant (e.g., we impose the phase value of the first SAR image equal to zero). Furthermore, in the more general case that the imposed perpendicular and temporal baseline constraints lead to the separation of the available SAR data set in more than one independent subset of small baseline SAR data pairs, the nonlinear minimization procedure can be independently performed for each subset. Accordingly, if we assume to face with L independent small baseline subsets of SAR acquisitions, it is easy to recognize that the unknown phase terms in (2a) and (2b) can be now estimated just up to arbitrary L additive constants, since only the phase differences between SAR acquisitions that belong to the same subset appear in (2a) and (2b). However, this does not represent a drawback of the algorithm because, once the noise-filtered interferograms belonging to each independent subset are retrieved, the application of the SBAS inversion allows us to compute the (unknown) $L-1$ phase offsets among the different L subsets [13].

Fourth, we highlight that, within the minimization scheme, we use only small perpendicular and temporal baseline interferograms because the large baseline ones, which are characterized by small values of the spatial coherence w_k [16], [17], are proved not contributing in (2a) and (2b) with relevant additional information. Indeed, we show in Appendix III the validity of this statement through a series of experimental tests carried out by applying the minimization procedure in (2a) and (2b) initially to the whole network of all of the possible $M^{(\max)} = (N+1)\cdot N/2$ interferograms and, subsequently, to limited groups of interferograms selected by progressively discarding the large baseline ones.

Finally, we stress that the proposed noise-filtering approach is intrinsically parallelizable by exploiting efficient concurrent and/or distributed processing approaches, thus allowing a significant reduction of the overall algorithm computational burden. This is due to the fact that the algorithm implements nonlinear minimization procedures that can be independently carried out for each pixel of the multi-look spatial scale.

B. Experimental Results

We investigated a data set consisting of 39 SAR images (Track 308, Frame 2754) collected by the ENVISAT sensor between December 2002 and August 2010 over the Abruzzi region. The test-site area includes the city of L'Aquila and its

[3]We remark that the aforementioned lack of triangularity has already been exploited, within the InSAR context, in order to develop the EMCF PhU algorithm [19], [48].

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surroundings, which were struck on April 6, 2009, by a 6.3-Mw earthquake that caused more than three hundred fatalities, thousands evacuees, and severe industrial and residential building damage [33]; the amplitude of a sample image of the investigated area is shown in Fig. 4(a).

The SAR images are represented in the $T \times B_{\perp}$ temporal/perpendicular baseline plane[4] as depicted in Fig. 4(b), while the corresponding set of small baseline interferograms is shown in Fig. 4(c). The latter has been obtained by imposing maximum perpendicular and temporal baseline values of 400 m and 2000 days, respectively. As a result, we retrieved a network of 338 differential SAR interferograms that were computed by performing a complex multi-look operation with 4 and 20 looks in the range and azimuth directions, respectively [38], [57], resulting in a pixel size of about 100 $\times$ 100 m. Moreover, within the interferogram [3], we used precise satellite orbit information and a three-arcsecond Shuttle Radar Topography Mission (SRTM) digital elevation model (DEM) of the region to remove the topographic phase contributions. Finally, the multi-look interferograms were also prefiltered by applying the approach described in [46].

To investigate the performance of the proposed approach, we first applied the nonlinear minimization procedure in (2a) and (2b) to the sequence of the generated 338 (original) multi-look small baseline interferograms to retrieve the corresponding sequence of the noise-filtered ones. Then, we computed, by applying (3), the two sequences of spatial coherence maps relevant to the original and noise-filtered interferogram sequences[5] and exploited them to identify, for each interferogram, the coherent pixels by imposing a threshold on the minimum allowed spatial coherence value (we used in our experiments a coherence threshold equal to 0.7). In Fig. 5(a) and (b), we portray the difference of the coherent pixel number between the noise-filtered and original interferograms plotted with respect to their perpendicular and temporal baselines, respectively. The inspection of Fig. 5(a) and (b) reveals that the coherent pixel number greatly increases when we pass from the original to the reconstructed interferograms (see the diamonds above the red line), except for a very limited group (about 10%) of reconstructed interferograms that exhibit a decreased number of coherent pixels (see the diamonds below the red line). Indeed, this happens in correspondence to interferograms with very small baselines that were (originally) extremely coherent and that play the key role to "inject" coherence in the noisy ones. To recover the high-quality coherence values of these interferograms, we simply apply to the computed noise-filtered interferograms the average operation stated in (4). As a result, we retrieved a new set of (average) noise-filtered interferograms that are characterized by generally improved coherence values. This is clearly visible in Fig. 5(c) and (d), where we portray the difference of the coherent pixel number between the average noise-filtered and original interferograms plotted with respect to their perpendicular and temporal baselines, respectively. In this case, negative values for all of the computed differences

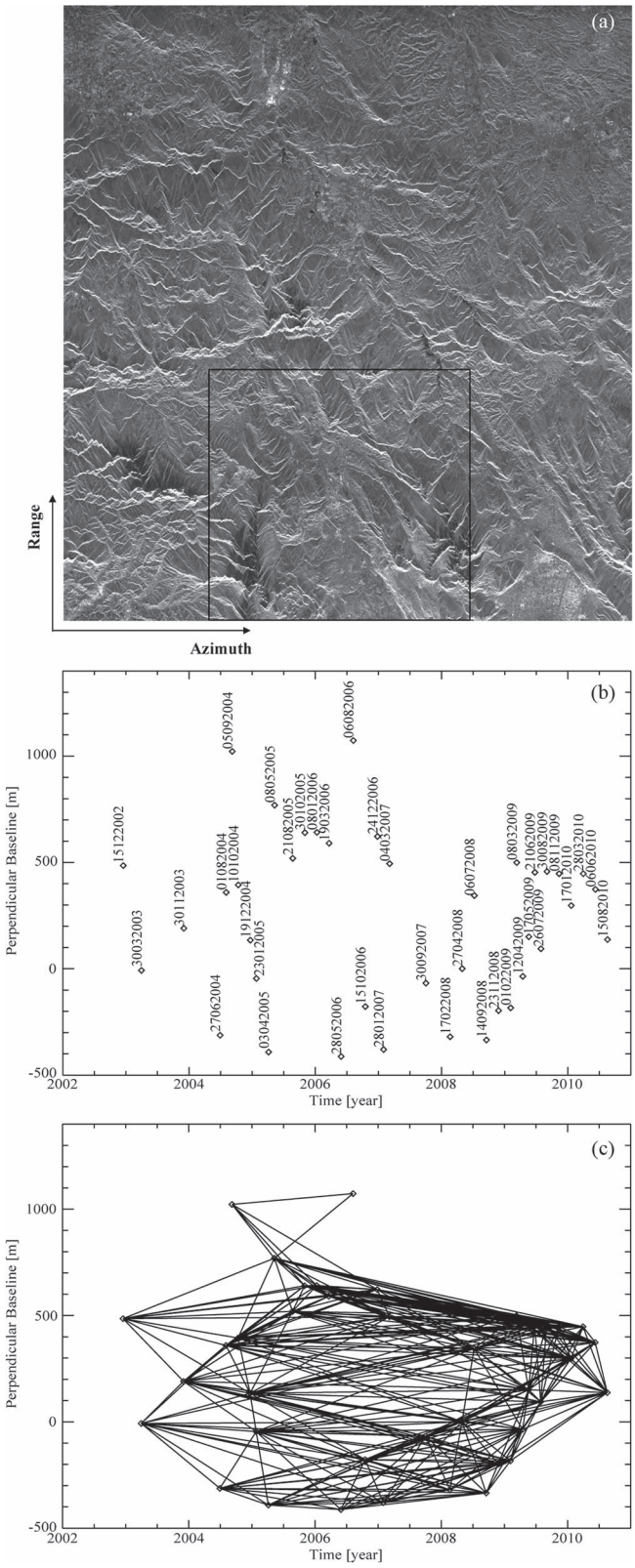


Fig. 4. Abruzzi region SAR data set. (a) SAR amplitude image relevant to the April 27, 2008, acquisition. The black box identifies the area nearby L'Aquila city that is particularly affected by the deformation caused by the April 6, 2009, earthquake. (b) Representation of the SAR data in the temporal/perpendicular baseline plane. (c) Distribution of the small baseline SAR data pairs used for the implementation of the proposed noise-filtering procedure. Note that they have been identified by imposing maximum perpendicular and temporal baseline values of 400 m and 2000 days, respectively, in the temporal/perpendicular baseline plane.

[4]To this aim, we must define the ratio between the axis units of the $T \times B_{\perp}$ plane, which is set in our analysis, similarly to [19], equal to $\delta T \delta b_{\perp}$, where δT and $\delta b_{\perp}$ represent the maximum allowed perpendicular and temporal baseline values.

[5]Obviously, in the case of noise-filtered interferograms, we replaced $\Delta\Phi$ with $\Delta\hat{\Phi}$ in (3).

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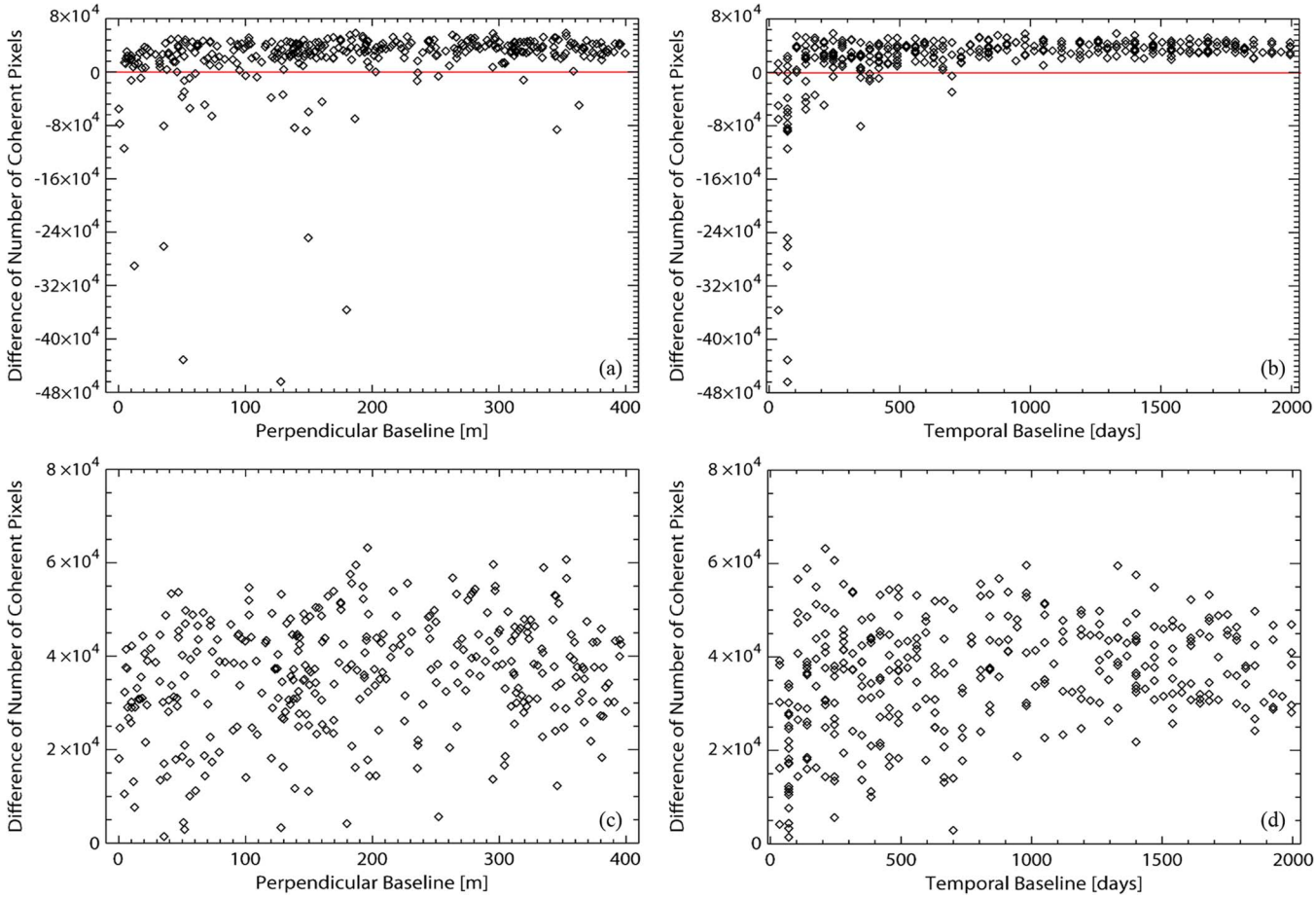


Fig. 5. Performance of the implemented interferogram noise-filtering procedure. (a) and (b) Difference of the coherent pixel number between the noise-filtered and original interferograms plotted with respect to their perpendicular and temporal baselines, respectively. (c) and (d) Same as (a) and (b) but the difference of the coherent pixel number is computed between the average noise-filtered and original interferograms.

are not found, thus implying that improved coherence levels characterize all of the retrieved interferograms.

To stress the interferogram quality improvement obtained by applying the proposed noise-filtering procedure, we show in Fig. 6 the original (left side) and corresponding average noise-filtered (right side) multi-look interferograms relevant to three SAR data pairs. By comparing the homologous interferograms of Fig. 6, we note that the decorrelation effects, clearly visible in Fig. 6(a)–(c), are drastically mitigated following the noise-filtering operation [see Fig. 6(d)–(f)]; this makes fully evident the effectiveness of the proposed algorithm. We also remark that the set of SB multi-look interferograms can be also prefiltered by using the filtering techniques proposed in [46] and [47]. The use of such prefiltering operations allows us to profitably increase the performance of the proposed noise-filtering approach. In particular, we show in Fig. 7 the effect of the proposed noise-filtering technique [Fig. 6(c) and (d)] when applied to the "original" interferogram [Fig. 7(a)] and to the corresponding prefiltered interferogram [Fig. 7(b)] retrieved by using the method proposed in [46]. The improvement that we obtain when we use as input of our noise-filtering technique a set of prefiltered interferograms rather than the corresponding set of "original" ones is evident.

With regard to the computational performance, we underline that the interferogram noise-filtering step was implemented through an algorithm running in parallel on several nodes. In particular, the Abruzzi data set processing required about 360 min by exploiting five 64-b processing nodes equipped with a dual quad-core AMD Opteron 2356 processor.

IV. Interferogram Selection Technique

We describe in this section the interferogram selection algorithm that we developed to identify the optimal network of small baseline interferograms, belonging to the set of previously noise-filtered small baseline interferograms,[6] to be exploited by the EMCF-SBAS processing chain. In particular, the identified sequence of interferograms represents the input of the EMCF step, which is typically based on the generation of spatial and temporal Delaunay triangulations. Moreover, equivalently to Section III, we also present the results achieved by applying the proposed interferogram selection algorithm to the ENVISAT SAR data set relevant to the Abruzzi region.

A. Algorithm

Let us assume the availability of a set of $N+1$ SAR images represented through a set of points, namely, $\mathbf{V} \equiv [V_0, V_1, \ldots, V_N]$, in the $T \times B_\perp$ plane. The generic arc connecting two

[6]For the sake of convenience, hereinafter, we identify as noise-filtered interferograms the ones retrieved by applying the algorithm discussed in Section III-A, i.e., the interferograms reconstructed following both the nonlinear minimization procedure in (2a) and (2b) and the average operation in (4).

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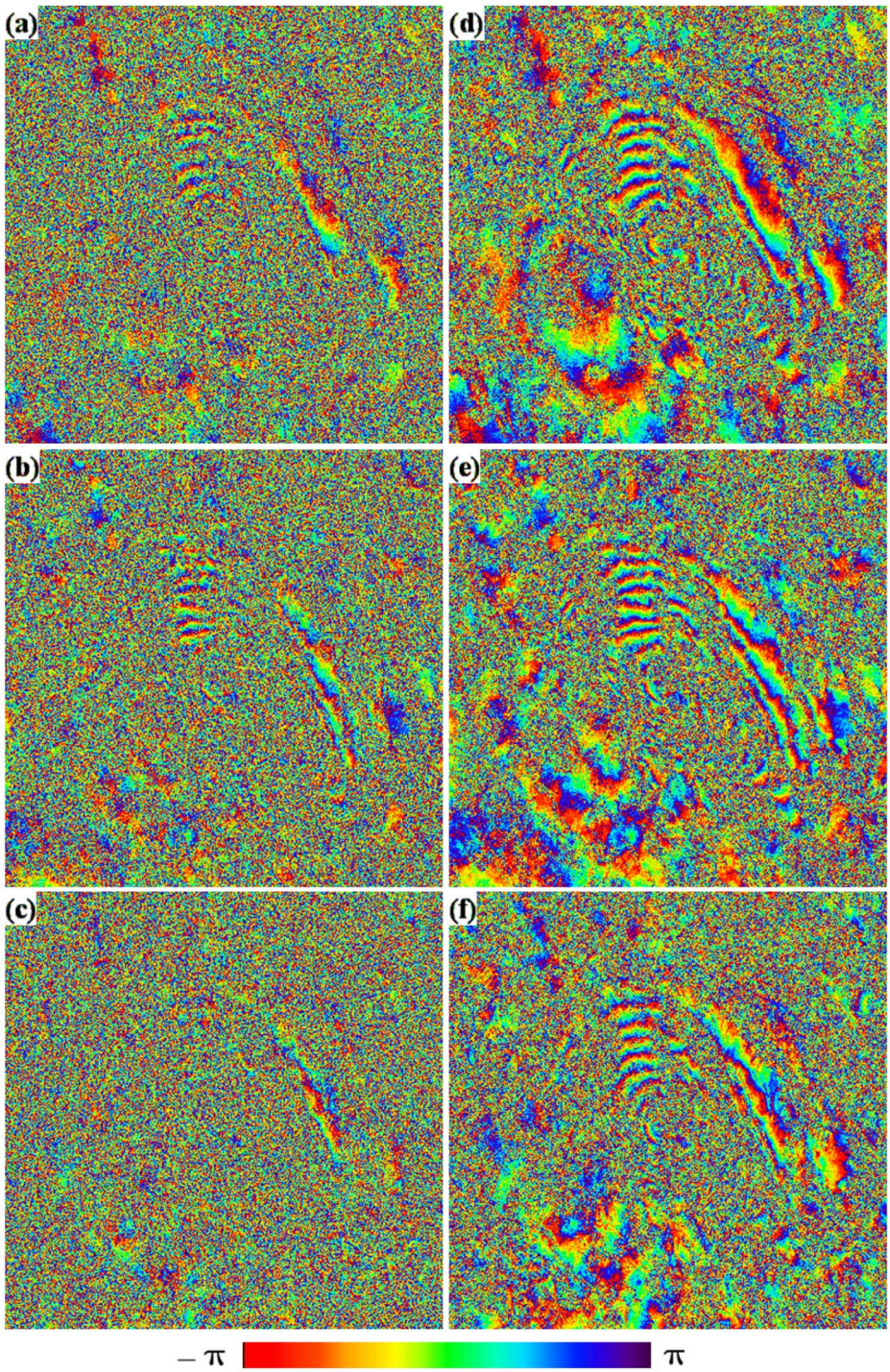


Fig. 6. Comparison between the original (left column) and noise-filtered (right column) multi-look interferograms relevant to the area of the Abruzzi region identified in Fig. 4(a) by the black box. (a)–(c) October 30, 2005, to November 8, 2009, August 21, 2005, to June 6, 2010, and August 1, 2004, to April 12, 2009, interferograms, characterized by perpendicular baseline values of 192, 145, and 395 m, respectively. (d)–(f) Noise-filtered multi-look interferograms corresponding to the ones in (a)–(c), respectively.

points in this plane and belonging to the temporal Delaunay triangulation, namely, $tr^D(\mathbf{V})$, allows us to identify the corresponding SAR data pair to be used for the interferogram generation. We observe that such a triangulation may involve arcs relevant to SAR data pairs characterized by large baseline values, thus potentially leading to operate with interferograms with significant decorrelation phenomena. Accordingly, to avoid these undesired effects without losing our triangulation representation, a straightforward solution is based on removing from $tr^D(\mathbf{V})$ all of the triangles involving arcs with too-large baselines, thus obtaining a corresponding reduced-Delaunay triangulation, namely, $tr^{rD}(\mathbf{V}) \subseteq tr^D(\mathbf{V})$. The identified reduced-Delaunay triangulation consists of M^{rD} small baseline interferograms (with $M^{rD} < M$), e.g., $\widehat{\Delta\Phi}^{rD} \equiv [\widehat{\Delta\Phi}_1, \widehat{\Delta\Phi}_2, \ldots, \widehat{\Delta\Phi}_{M^{rD}}]$, that, in our case, belong

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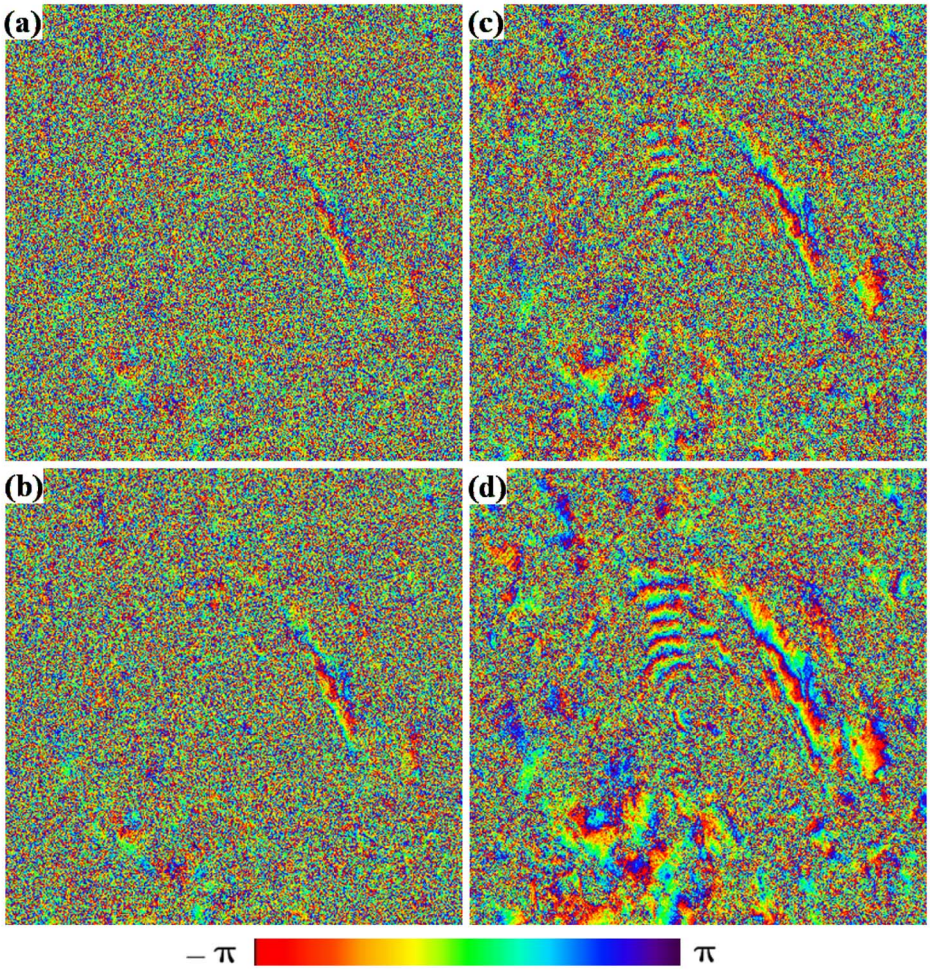


Fig. 7. Impact of the prefiltering operation on the proposed interferogram noise-filtering technique for the August 1, 2004, to April 12, 2009, interferometric SAR data pair. (a) Original (noisy) interferogram. (b) Prefiltered interferogram obtained by using the technique proposed in [46]. (c) Noise-filtered interferogram using the proposed noise-filtering method starting from (a). (d) Same as (c) but starting from (b).

to the set of noise-filtered small baseline interferograms ($\Delta\widehat{\Phi} \equiv [\Delta\widehat{\Phi}_1, \Delta\widehat{\Phi}_2, \ldots, \Delta\widehat{\Phi}_M]$) retrieved by applying the procedure reported in Section III-A.

The Delaunay spatial triangulation is, on the other hand, built in the $A_Z \times R_G$ azimuth/range spatial plane and involves the set of coherent pixels common to the small baseline interferograms belonging to $tr^{rD}(\mathbf{V})$. Such coherent pixels are identified following the same strategy adopted in [21] that is based on analyzing the interferometric phase information of the M^{rD} noise-filtered interferograms $\Delta\widehat{\Phi}^{rD}$. In particular, we first estimate the following (complex) factor:

$$\begin{aligned}\xi_k(x,r) &= \frac{1}{(2N_A+1)(2N_R+1)} \sum_{h=-N_A}^{N_A} \\ &\quad \times \sum_{p=-N_R}^{N_R} \exp\left[j \cdot \Delta\widehat{\Phi}_k^{rD}(x+h, r+p)\right] \\ &= w_k^{rD}(x,r) \exp\left[j\Delta\widehat{\Phi}_k^{rD(LP)}(x,r)\right] \\ &\quad \forall\, k = 1, \ldots, M^{rD} \end{aligned} \tag{5}$$

with $2N_A + 1$ and $2N_R + 1$ being the number of azimuth and range pixels within the used boxcar averaging window, which is centered around the generic pixel P of radar coordinates (x, r). Moreover, the amplitude term $w_k^{rD}(\cdot)$[7] and the phase factor $\Delta\widehat{\Phi}_k^{rD(LP)}$ provide an estimate of the spatial coherence (directly computed from the interferogram phase) and of the spatial low-pass phase component of the kth interferogram, respectively.

Then, we compute the values of the following factor [21]:

$$\Omega^{rD}(P) = \frac{\left|\sum_{k=1}^{M^{rD}} \exp\left[j\left(\Delta\widehat{\Phi}_k^{rD}(P) - \Delta\widehat{\Phi}_k^{rD(LP)}(P)\right)\right]\right|}{M^{rD}} \quad \forall\, P \in A_Z \times R_G \tag{6}$$

that may be considered as an "equivalent spatial coherence" and is used to identify the set of coherent pixels of the scene: $C \equiv \{P : \Omega^{rD}(P) \geqslant \omega\}$ to be further analyzed, where ω is a proper threshold that is set equal to 0.5 in the following experiments.

The proposed interferogram selection procedure is aimed at identifying the optimal reduced triangulation $t\tilde{r}^r(\mathbf{V})$ in the $T \times B_\perp$ plane to be effectively used by the EMCF algorithm,

[7]The w_k^{rD} term in (5) is the same as in (3), but now, it is computed on the set of M^{rD} SB interferograms.

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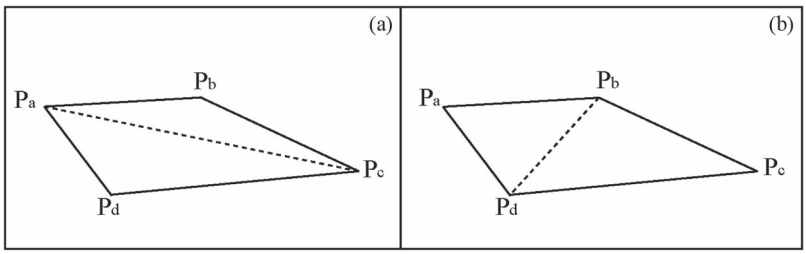


Fig. 8. Example of the "edge flip" operation carried out on the $P_aP_bP_cP_d$ use quadrilateral. The P_a–P_c diagonal in (a) is replaced by the P_b–P_d diagonal in (b).

in place of the reduced-Delaunay triangulation $tr^{rD}(\mathbf{V})$ that may still involve some decorrelated small baseline interferograms. In particular, our algorithm implements a searching procedure based on the simulated annealing (SA) technique [44] that is a probabilistic method consisting in a discrete-time inhomogeneous Markov chain, e.g., $x(t)$, where t is the chain evolution time. Note that the SA approach allows finding the global minimum of a cost function that may possess several local minima by emulating the physical process whereby a solid is slowly cooled until its structure is eventually frozen [44].

The implemented SA-based approach identifies as the optimal reduced triangulation $t\tilde{r}^r(\mathbf{V})$ the one that maximizes the average values of the equivalent coherence [see (6)], relevant to the set of small baseline multi-look interferograms. It is computed over the whole scene (or over a portion of it), e.g., S, consisting in N_P pixels, and is represented as follows:

$$t\tilde{r}^r = \arg\max\left\{\eta\left[tr^r(V)\right]\right\} \tag{7a}$$

$$\eta\left[tr^r(V)\right] = 1/N_P \sum_{P\in S} \Omega^r(P) = \frac{1}{N_P}\sum_{P\in S}\frac{\left|\sum_{k=1}^{M^r}\exp\left[j\left(\Delta\widehat{\Phi}_k^r(P)-\Delta\widehat{\Phi}_k^{r(LP)}(P)\right)\right]\right|}{M^r} \tag{7b}$$

where $tr^r(\mathbf{V})$ is the reduced triangulation identified by removing, in the generic (full) triangulation $tr(\mathbf{V})$, the triangles involving at least one large baseline interferogram and $\Omega^r(\cdot)$ is the equivalent coherence computed as in (6) by considering the interferograms belonging to $tr^r(\mathbf{V})$, M^r is the number of small baseline interferograms belonging to $tr^r(\mathbf{V})$, and $\Delta\widehat{\Phi}_k^r$ and $\Delta\widehat{\Phi}_k^{r(LP)}$, $\forall\, k = 1,\ldots,M^r$ are the considered noise-filtered interferograms and their spatial low-pass phase components, respectively.

We remark that the problem stated in (7a) and (7b) is equivalent to the following one:

$$t\tilde{r} = \arg\min\left\{\chi\left[tr^r(V)\right]\right\} \tag{8a}$$

$$\chi\left[tr^r(V)\right] = 1 - \eta\left[tr^r(V)\right] \tag{8b}$$

where χ represents the cost function to be minimized by the SA algorithm.

To solve the problem (8a) and (8b), we select as initial state for the SA procedure the full Delaunay triangulation relevant to the set of vertices $\mathbf{V}$: $x(t)|_{t=0} \equiv tr^D(\mathbf{V})$ to which corresponds a cost function value $\chi[tr^{rD}(V)]$, where, as already stated earlier, $tr^{rD}(\mathbf{V})$ represents the reduced-Delaunay triangulation corresponding to $tr^D(\mathbf{V})$. Starting from this state, at each step, the algorithm mutates its current state $x(t)$ (corresponding to a generic triangulation $tr(\mathbf{V})$) with a certain probability $pr(t)$ by performing a conventional topological operation on the network of interferograms that is usually referred to as edge flip [58]. In particular, this operation consists in randomly selecting an inner edge of the current triangulation $tr(\mathbf{V})$ and replacing it with the other diagonal of the quadrilateral that surrounds the edge (see Fig. 8). This is done only if the quadrilateral is convex and the replaced edge corresponds to a small baseline SAR data pair. Conversely, the new triangulation is considered illegal, and the SA state does not change.

The probability $pr(t)$ to accept at time t a new triangulation is computed as follows:

$$pr(t) = \exp\left\{-\frac{1}{T(t)}\max[0, \chi_{\text{new}} - \chi_{\text{curr}}]\right\} \tag{9}$$

where χ_{new} and χ_{curr} are the cost function values [see (8a) and (8b)] relevant to the new and current states (triangulations), respectively, and $T(t)$ is the so-called temperature at time t.[8] When the new triangulation is rejected, the new SA state is set equal to the previous one, and the triangulation does not change. The algorithm goes on until the convergence condition is met, i.e., the SA state remains unchanged within a fixed number of iterations, namely, NI.

As a result of the SA-based procedure, we retrieve, among the whole set of possible reduced triangulations $\{tr^r(V)\}$, the optimal one $t\tilde{r}^r(V)$ that finally leads to the identification of the interferograms to be exploited by the cascade of the EMCF and SBAS steps.

We observe that the use of the proposed interferogram selection procedure may create disconnected reduced triangulations within the whole set of SB interferograms; this, however, does not represent a critical issue for the subsequent PhU step. Indeed, the identified interferograms, which may potentially belong to disconnected reduced triangulations, can be separately unwrapped by applying the EMCF procedure [19]. Moreover, the corresponding sets of unwrapped interferograms result independent of each other, but they can be easily "linked" through the SBAS inversion strategy [13]. Nonetheless, we remark that, for a relatively large amount of data, as for the case of the ENVISAT ones, the probability to obtain one single connected data set is very high.

[8]The temperature $T(t)$ usually decays with respect to time with a decay factor α that we set equal to 0.9 in the following experimental analysis.

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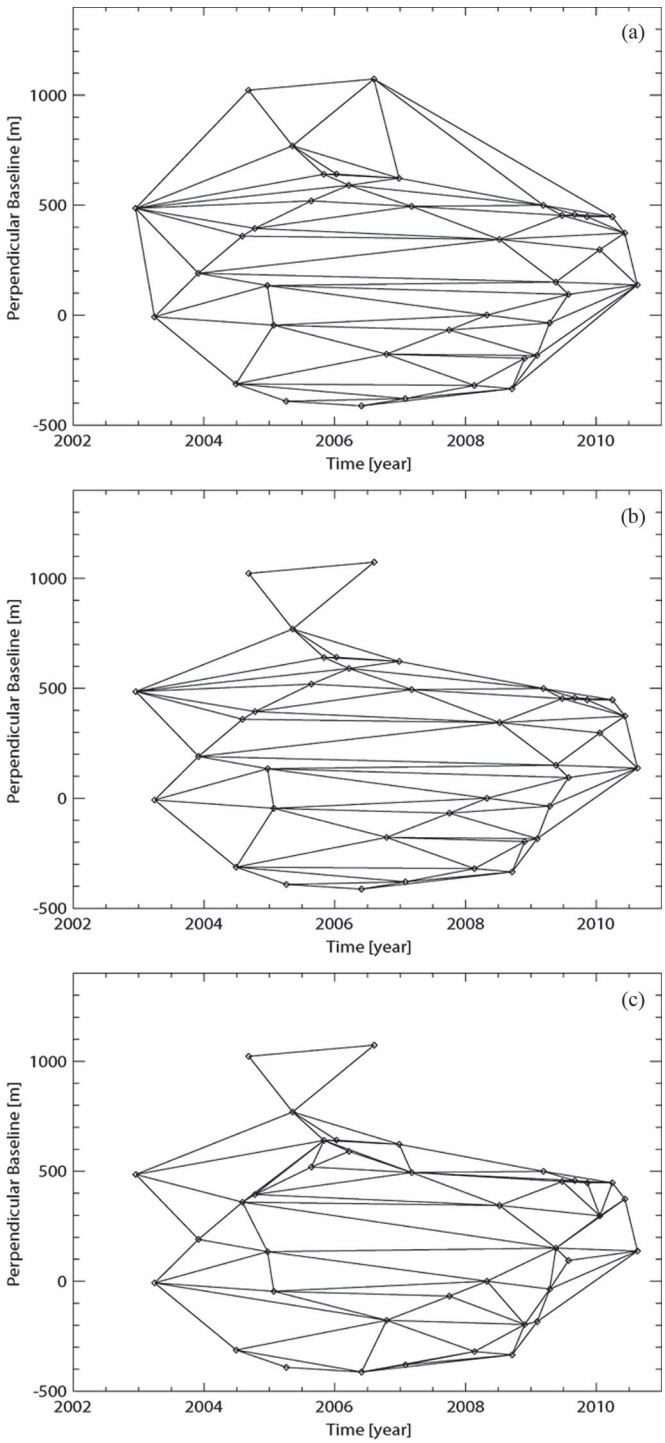


Fig. 9. SAR data representation in the temporal/perpendicular baseline plane for the Abruzzi region. (a) Delaunay triangulation. (b) Reduced-Delaunay triangulation after the removal of triangles with sides characterized by perpendicular and temporal baseline values exceeding the selected thresholds (corresponding, in our experiments, to 400 m and 2000 days, respectively). (c) Optimal reduced triangulation obtained by applying the implemented interferogram selection procedure.

B. Experimental Results

To illustrate the performance of the proposed interferogram selection procedure, we exploited the same ENVISAT SAR data set relevant to the Abruzzi region used for the experimental results described in Section III-B.

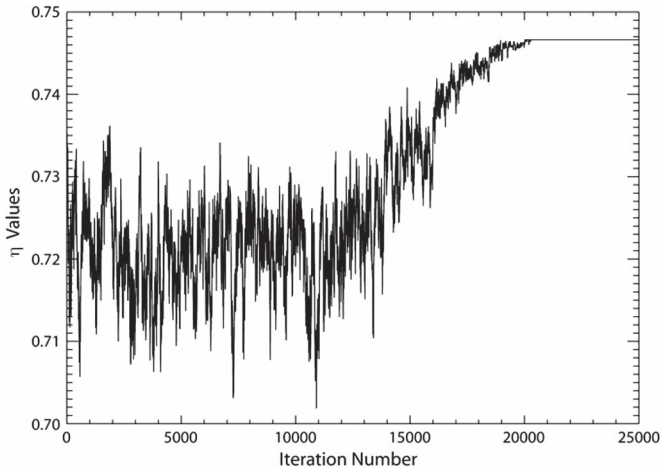


Fig. 10. Variation of the η values with respect to the number of iterations performed during the SA-based procedure.

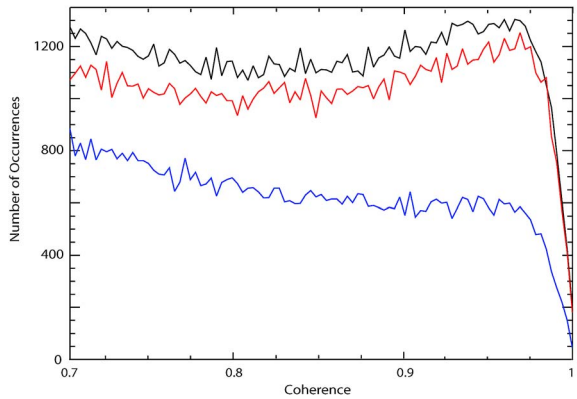


Fig. 11. Histograms of the equivalent coherence maps relevant to the Abruzzi region computed by considering the original interferograms and the reduced-Delaunay triangulation (blue line), the noise-filtered interferograms and the reduced-Delaunay triangulation (red line), and the noise-filtered interferograms and the optimal reduced triangulation (black line).

Starting from the available set of 39 SAR images, we first generated the corresponding (full) Delaunay triangulation in the $T \times B_{\perp}$ plane [see Fig. 9(a)], from which we discarded all of the triangles involving at least one "large baseline" interferogram (by imposing the typical maximum perpendicular and temporal baseline values of 400 m and 2000 days, respectively), thus achieving the reduced triangulation shown in Fig. 9(b). Then, we applied the SA-based searching procedure described in the previous Section IV-A, by setting $T(t)|_{t=0} = 100$ and $NI = 30$. Fig. 10 shows the variation of the η values with respect to the number of iterations performed during the SA-based procedure, leading to the final solution, which corresponds to the optimal reduced triangulation depicted in Fig. 9(c).

To demonstrate the effectiveness of the proposed approach, we computed the equivalent spatial coherence maps, obtained following the lines of (6), by considering the sequences of the multi-look small baseline original and noise-filtered interferograms relevant to the reduced-Delaunay triangulation of Fig. 9(b) and the sequence of the multi-look noise-filtered interferograms corresponding to the reduced "optimal" triangulation depicted in Fig. 9(c). The histograms of the three equivalent coherence maps, which are shown in Fig. 11, reveal a general

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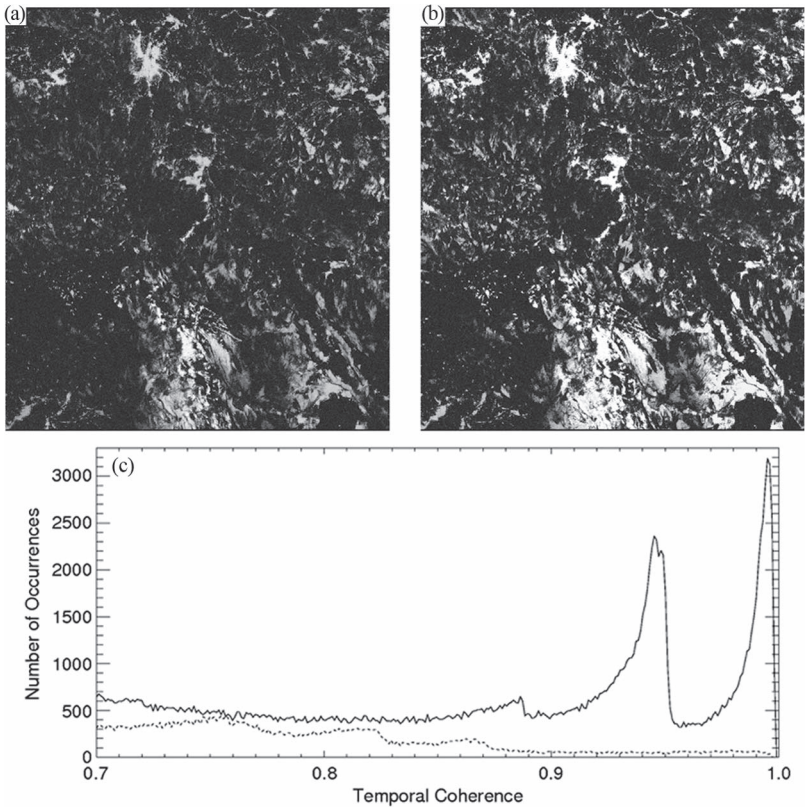


Fig. 12. Comparison of the temporal coherence maps obtained by using the (a) original and (b) advanced EMCF-SBAS processing chains, respectively. (c) Histograms of the corresponding coherence maps: the dotted line refers to the original EMCF-SBAS results, and the continuous line refers to the advanced EMCF-SBAS ones.

increase of the coherence values. In particular, we observe that the number of coherent pixels obtained by considering the reduced-Delaunay triangulation increases by about 50% as we move from the original (blue line) to the noise-filtered (red line) interferograms. The improvement is even more pronounced as we estimate the equivalent coherence of the noise-filtered interferograms over the optimal reduced triangulation (black line); in this case, we obtain an increase of about 60%. As a further remark, we also underline that the interferogram selection step only requires a minimal increase of the computational load. In fact, on a single 64-b processing node equipped with a dual quad-core AMD Opteron 2356 processor, the procedure required about 40 min to select the ENVISAT interferograms of the Abruzzi test-site.

V. Performance Analysis of the Advanced EMCF-SBAS DInSAR Processing Chain

We extensively investigate in this section the impact of the two new preprocessing steps, discussed in Sections III and IV, on the overall performance of the EMCF-SBAS processing chain. To this aim, we compare the DInSAR results achieved by applying the original (see Fig. 1) and advanced (see Fig. 2) EMCF-SBAS procedures. In particular, our analysis starts by considering the ENVISAT SAR data set acquired over the Abruzzi region, already exploited for the experiments shown in Sections III and IV. Subsequently, it is extended by taking into account two additional ENVISAT data sets relevant to Mt. Etna volcano (Southern Italy) and Yellowstone Caldera (WY, USA).

A. Advanced EMCF-SBAS Analysis: The Abruzzi Region Case Study

Let us start our analysis by focusing on the Abruzzi region SAR data set [see Fig. 4(a)]. In particular, we first applied the original EMCF-SBAS processing chain to the set of original multi-look small baseline interferograms belonging to the reduced-Delaunay triangulation [see Fig. 7(b)]; then, we applied the advanced EMCF-SBAS processing chain to the set of noise-filtered multi-look small baseline interferograms relevant to the optimal reduced triangulation depicted in Fig. 7(c). As a first result, we show in Fig. 12(a) and (b) the EMCF-SBAS temporal coherence maps [19] achieved by applying the original and advanced processing chains, respectively. The coherence improvement is evident as we move from the original to the advanced EMCF-SBAS maps; this is even more evident by comparing the two temporal coherence histograms plotted in Fig. 12(c). Indeed, we observe an increase of about 350% of the pixel number with a temporal coherence value greater than 0.7 (which is a typical value used in EMCF-SBAS applications [19]). Fig. 13(a) and (b) reports the mean deformation velocity maps (color) of the Abruzzi area as computed by using the original and advanced EMCF-SBAS processing chains, respectively. The retrieved velocity maps have been superimposed on an amplitude multi-look SAR image (gray scale) of the study area, wherein the zones exhibiting low values of the temporal coherence factors have properly been masked out. Note that this kind of representation is particularly effective because it provides direct information on the estimated mean

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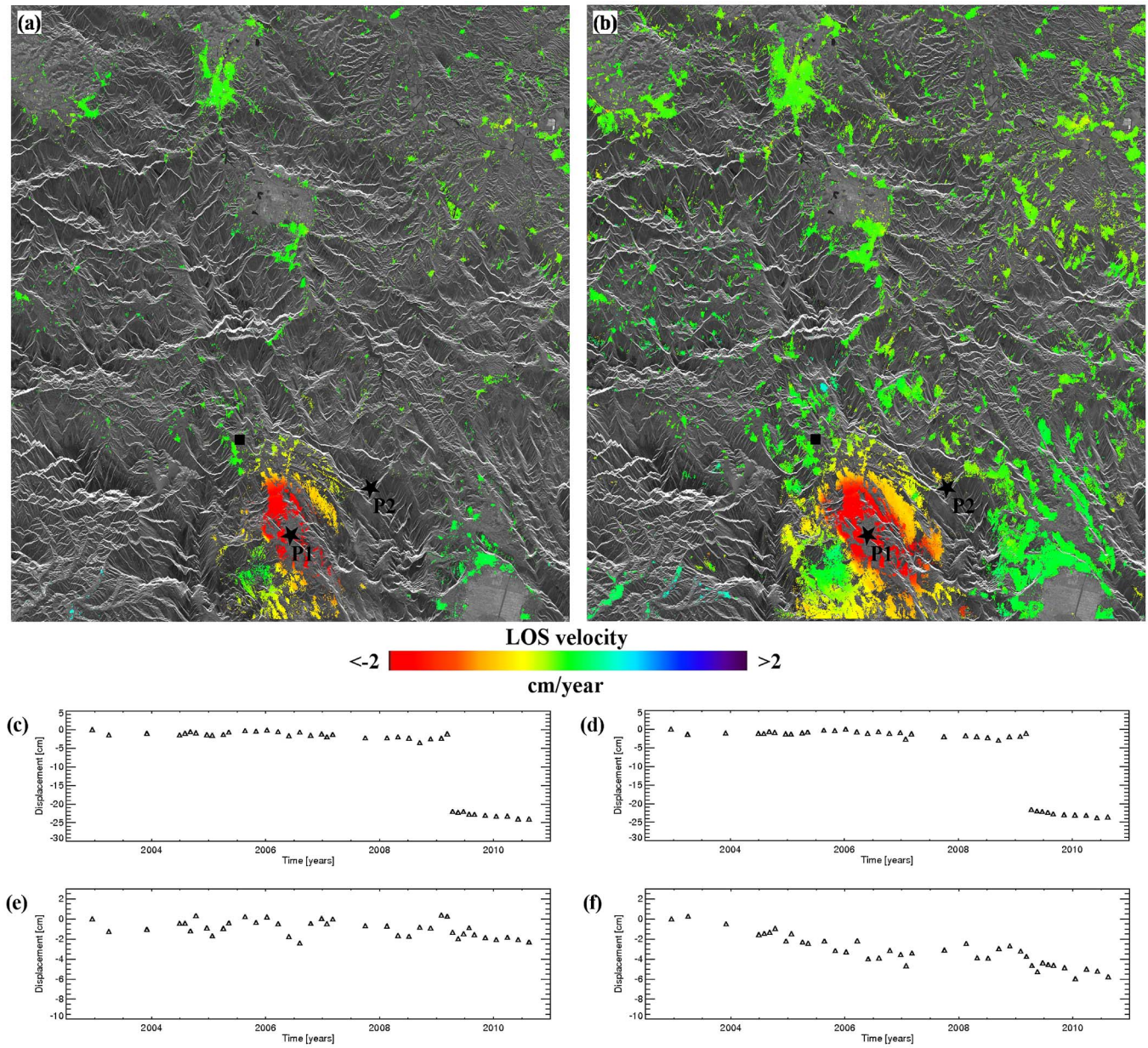


Fig. 13. Mean deformation velocity maps (in color) of the Abruzzi region, computed in coherent pixels only and superimposed on the SAR amplitude image (gray-scale representation) of the zone, retrieved by applying the (a) original and (b) advanced EMCF-SBAS processing chains, respectively. The black square marks the location of the reference SAR pixel. (c) and (d) Deformation time-series relevant to a pixel located in the area of the maximum displacement labeled as P1 in (a) and (b), retrieved by applying the (c) original and (d) advanced (d) EMCF-SBAS processing chain. (e) and (f) Same as (c) and (d) but for the pixel labeled as P2 in (a) and (b).

deformation rate; note also that all of the measurements are evaluated with respect to an assumed stable pixel whose location is identified with a black square in Fig. 13(a) and (b). The comparison of the two velocity maps clearly demonstrates the impact that the advanced EMCF-SBAS processing chain has on the retrieved deformation products. Indeed, although both maps reveal a spatially extended deformation pattern associated to the earthquake that struck the L'Aquila city and its surroundings in 2009, the one reported in Fig. 12(b) is significantly spatially denser, thus allowing a more detailed analysis of the observed phenomena. Moreover, we report in Fig. 13(c) and (d) the displacement time-series relevant to the pixel labeled as P1 in Fig. 13(a) and (b), which is located in the area of maximum coseismic LOS-projected deformation; both plots clearly show the deformation signal associated to the occurred seismic event [33]. In Fig. 13(e) and (f), we report the deformation time-series for the two pixels labeled as P2 in Fig. 13(a) and (b), located in an area that can be investigated by the advanced EMCF-SBAS processing chain only. From the plot inspection, it is clear that no significant differences are found in the deformation time-series for those pixels located in the areas where the coherent pixel density is high, also without applying the advanced processing chain [as for the pixel P1 in Fig. 13(a) and (b)]. Conversely, relevant differences are detected in those areas where the advanced EMCF-SBAS processing chain has increased the number of coherent pixels [as for the pixel P2 in

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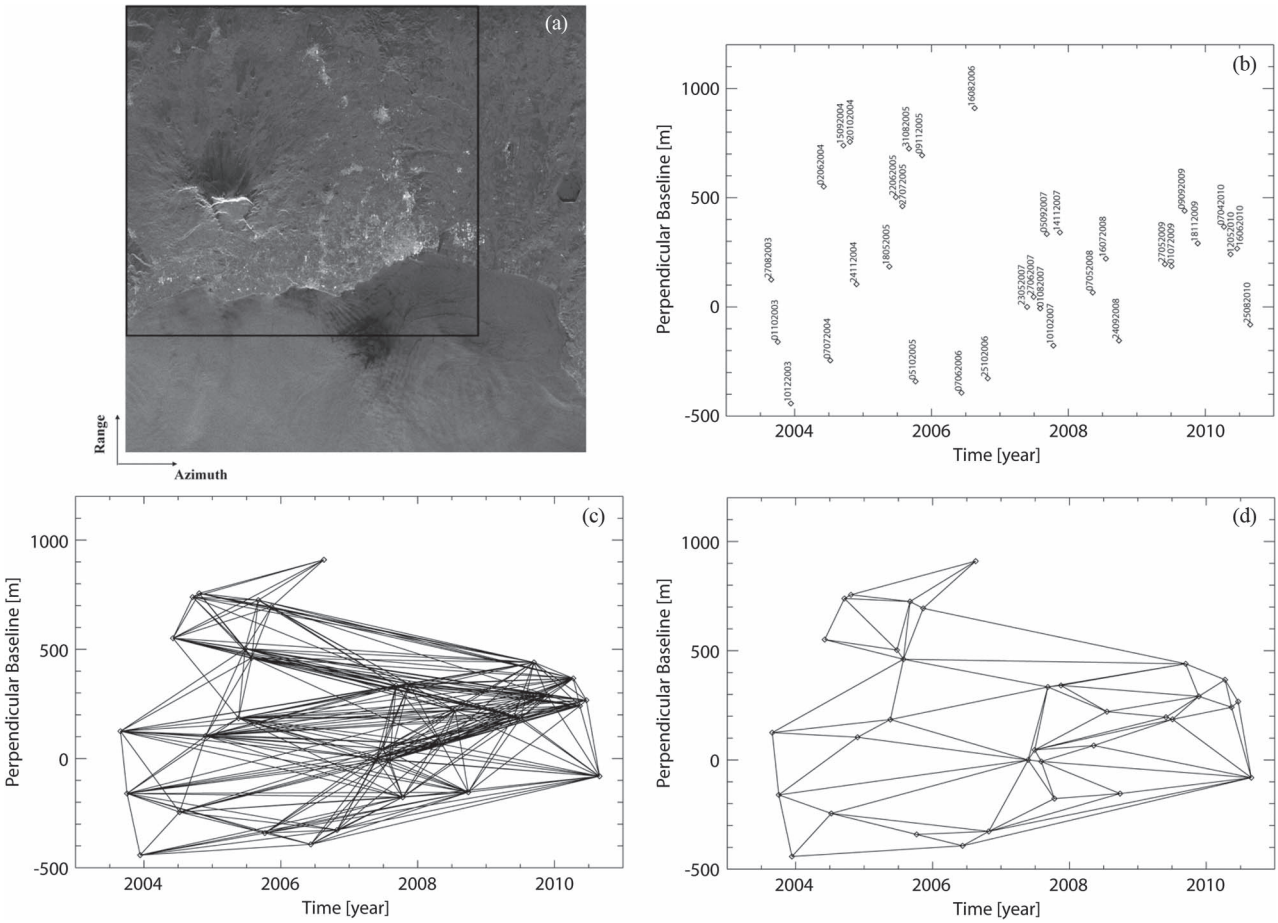


Fig. 14. Mt. Etna volcano SAR data set. (a) SAR amplitude image relevant to the May 23, 2007, acquisition. The black box identifies the summit and flank volcano. (b) Representation of the SAR data in the temporal/perpendicular baseline plane. (c) Distribution of the small baseline SAR data pairs used for the implementation of the proposed noise-filtering procedure. Note that they have been identified by imposing maximum spatial and temporal baseline values of 400 m and 2000 days, respectively, in the temporal/perpendicular baseline plane. (d) Optimal reduced triangulation obtained after the application of the proposed interferogram selection procedure.

Fig. 13(a) and (b)]. This finding clearly emphasizes the impact on the final result of the proposed advanced EMCF-SBAS processing chain, and above all, it shows that the presented advanced processing chain is extremely robust, also in the presence of strongly nonlinear deformation, as for the Abruzzi case study.

B. Advanced EMCF-SBAS Analysis: The Mt. Etna Volcano Case Study

The second case study is represented by Mt. Etna, one of the most active volcanoes in the world, for which an archive of 34 ENVISAT SAR images (Track 222, Frame 2853) collected between August 2003 and August 2010 was exploited; the amplitude of a sample image of the investigated area is shown in Fig. 14(a). The SAR data distribution in the $T \times B_{\perp}$ plane is depicted in Fig. 14(b), while the corresponding set of small baseline interferograms, obtained by imposing maximum perpendicular and temporal baseline values of 400 m and 2000 days, respectively, is shown in Fig. 14(c). As a result, we retrieved a network of 288 small baseline interferograms that were computed, as for the Abruzzi region case study, by performing complex multi-look (with 4 and 20 looks in the range and azimuth directions, respectively [38], [57]) and prefiltering (by using the approach described in [46]) operations.

We applied the advanced EMCF-SBAS processing chain to this sequence of interferograms. Accordingly, we first retrieved through the noise-filtering procedure (described in Section III) the corresponding sequence of noise-filtered interferograms. Just to illustrate the effectiveness of the noise-filtering algorithm, we show in Fig. 15 a selection of original (left side) and noise-filtered (right side) interferograms; the comparison of the homologous interferograms reveals a very significant coherence improvement, which is particularly evident in correspondence to the summit and seaside slopes of the volcano. Subsequently, we applied the interferogram selection procedure (reported in Section IV) to the computed noise-filtered interferograms; this allowed us to identify the optimal reduced triangulation of small baseline multi-look interferograms depicted in Fig. 14(d), representing the input of the EMCF-SBAS processing chain.

As a final result, we retrieved the displacement time-series and the mean deformation velocity map of the study area in correspondence to the pixels characterized by temporal coherence values greater than 0.7. To investigate the deformation product quality improvement gained owing to the application of the advanced EMCF-SBAS processing chain, we compared

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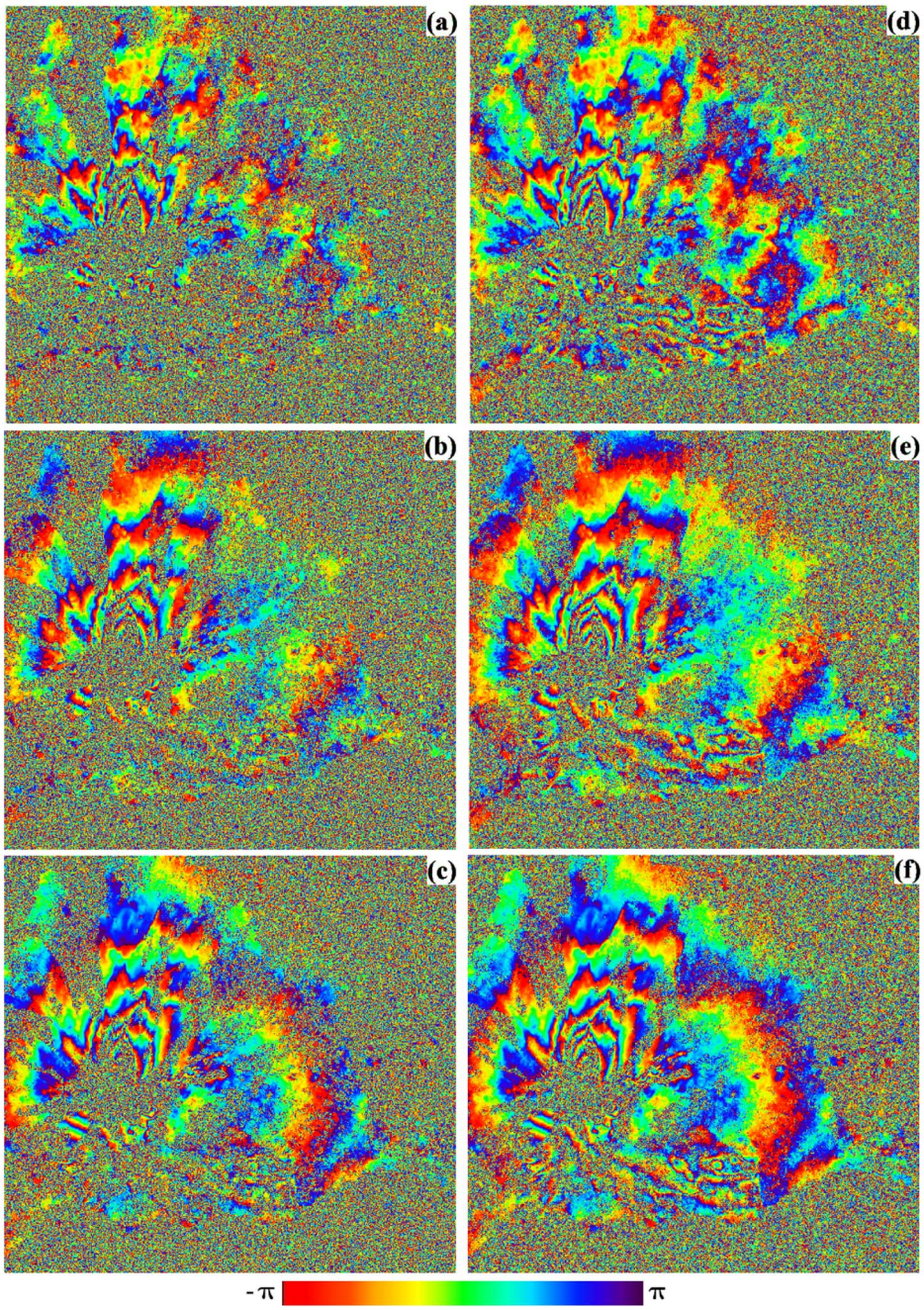


Fig. 15. Comparison between original (left column) and noise-filtered (right column) multi-look interferograms relevant to the area of the Mt. Etna volcano identified in Fig. 12(a) by the black box. (a)–(c) June 2, 2004, to July 1, 2009, November 9, 2005, to April 7, 2010, and November 24, 2004, to November 18, 2009, interferograms, characterized by perpendicular baseline values of 359, 326, and 185 m, respectively. (d)–(f) Noise-filtered multi-look interferograms corresponding to the ones in (a)– (c), respectively.

the computed mean deformation velocity map depicted in Fig. 16(b) to the one obtained by applying the original EMCF-SBAS processing chain and shown in Fig. 16(a). As for the Abruzzi region case study, the exploitation of the advanced EMCF-SBAS procedure has led to a considerable increase of the total number of coherent pixels, corresponding, in this case, to about 40%. The two velocity maps show a very complex deforming scenario characterizing Mt. Etna volcano. To provide evidences of the temporal evolution of the ongoing deformation phenomena, we present the retrieved displacement time-series computed in correspondence to the pixels labeled as P1, P2, P3, and P4 and identified via black stars in Fig. 16(b) in the plots of Fig. 16(c)–(f), respectively.

C. Advanced EMCF-SBAS Analysis: The Yellowstone Caldera Case Study

As a final experiment, we focus on the Yellowstone Caldera, representing one of the largest and most active volcanic systems in the world. We analyzed the temporal evolution of the surface deformation occurring in the area by applying the implemented advanced EMCF-SBAS processing chain to a

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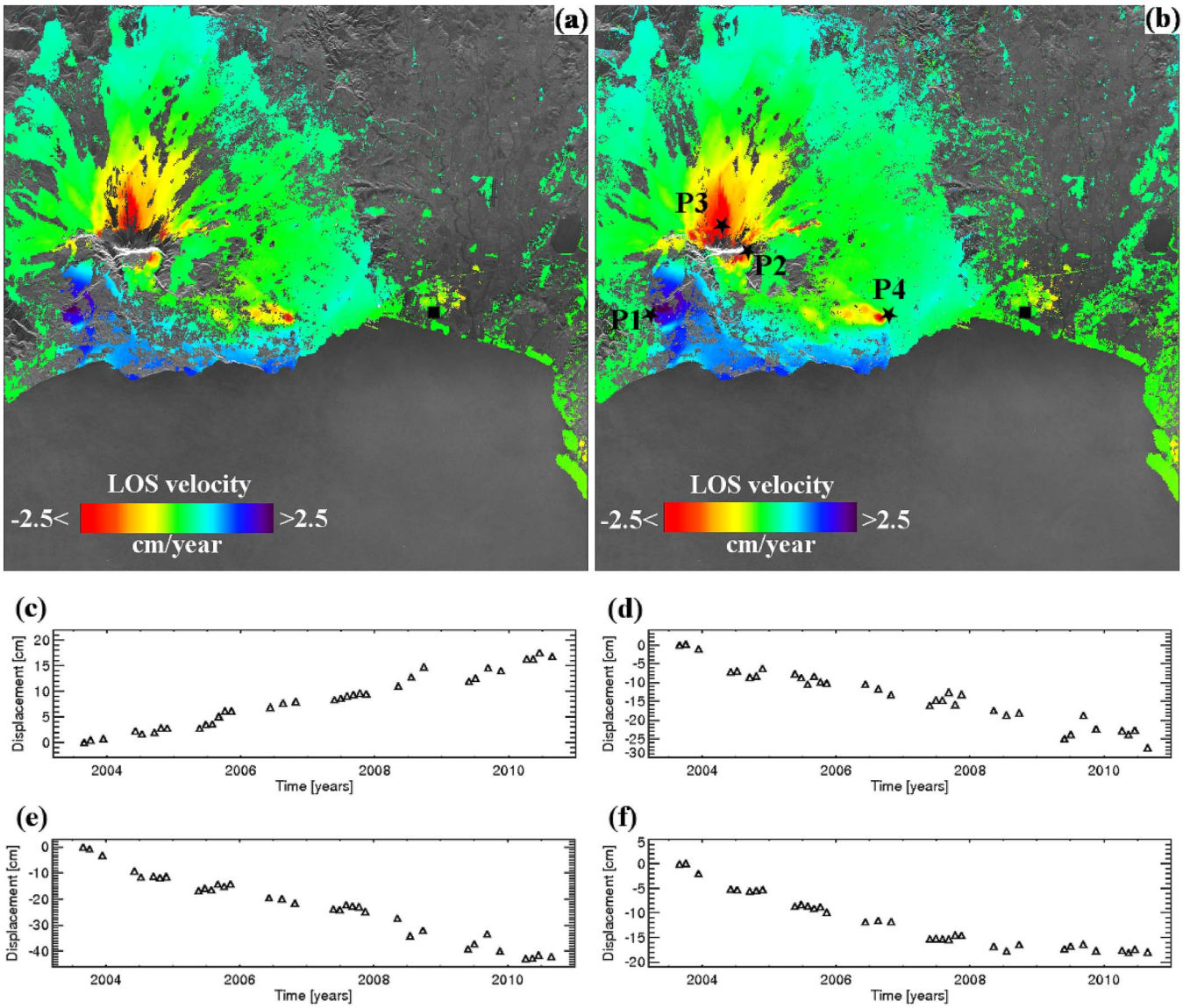


Fig. 16. Mean deformation velocity maps (in color) of Mt. Etna volcano, computed in coherent pixels only and superimposed on the SAR amplitude image (gray-scale representation) of the zone, retrieved by applying the (a) original and (b) advanced EMCF-SBAS processing chains, respectively. The black square marks the location of the reference SAR pixel. (c)–(f) Deformation time-series relevant to the four pixels identified via black stars in Fig. 14(b).

set of 22 ENVISAT images (Track 41, Frame 2709) acquired from May 2005 to September 2010. The SAR data distribution in the $T \times B_{\perp}$ plane is depicted in Fig. 17(a), whereas the corresponding set of 122 small baseline interferograms[9] is shown in Fig. 17(b); note that they have been selected, as for the two previous case studies, by imposing maximum perpendicular and temporal baseline values of 400 m and 2000 days, respectively.

As a first result, we present in Fig. 18 two noise-filtered interferograms (right side) that have been compared to the corresponding original ones (left side); the comparison of the homologous interferograms makes it evident the effectiveness of the noise-filtering procedure, which allows us to significantly improve the spatial coherence of the computed interferograms. Indeed, in Fig. 18(c) and (d), the fringes associated to the caldera activity are easily recognizable at variance with what are shown in Fig. 18(a) and (b). For the sake of brevity, in the following, we only present the results that we achieved by exploiting the advanced EMCF-SBAS processing chain without comparing them to the ones obtained by using the original EMCF-SBAS processing chain. Accordingly, we portray in Fig. 19(a) the retrieved advanced EMCF-SBAS mean deformation velocity map, which allows us to retrieve the very complex deformation scenario affecting the caldera region and its surroundings, where uplift effects and broad subsidence patterns characterize the detected deformation field. In addition, in Fig. 19(b)–(e), we report the deformation time-series relevant to four pixels whose locations are identified by the black stars labeled as P1, P2, P3, and P4 in Fig. 19(a).

[9]Also, in this case, the interferograms were computed by performing complex multi-look (with 4 and 20 looks in the range and azimuth directions, respectively) and prefiltering operations.

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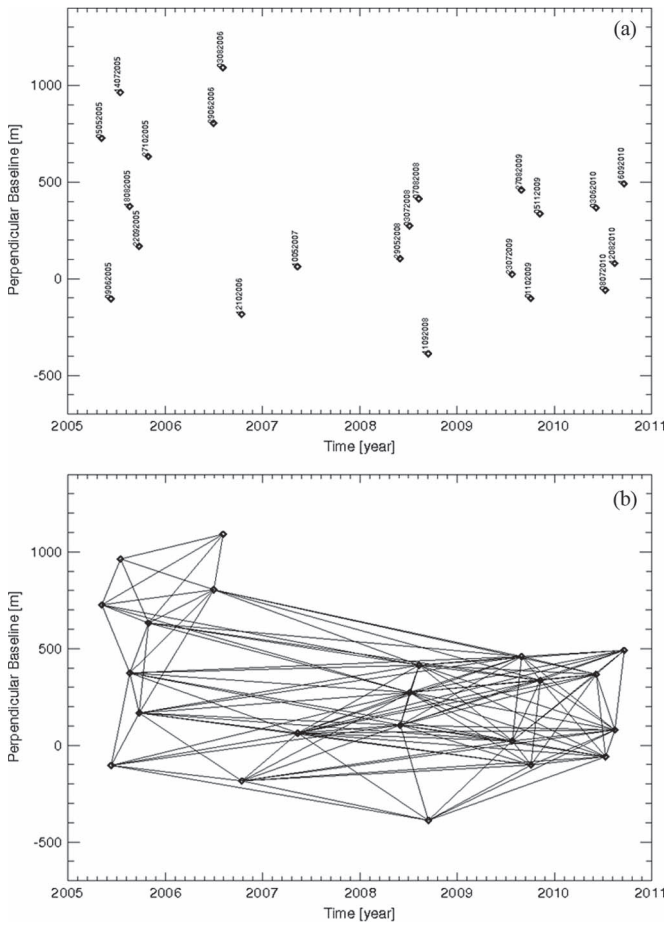


Fig. 17. Yellowstone Caldera SAR data set. (a) Representation of the SAR data in the temporal/perpendicular baseline plane. (b) Distribution of the small baseline SAR data pairs used for the implementation of the proposed noise-filtering procedure. Note that they have been identified by imposing maximum spatial and temporal baseline values of 400 m and 2000 days, respectively, in the temporal/perpendicular baseline plane.

VI. Conclusion

We have presented in this paper an advanced version of the EMCF-SBAS processing chain for the generation of ground deformation maps and corresponding time-series. The key elements of our approach are two additional preprocessing steps implementing an effective interferogram noise-filtering technique and an optimal interferogram selection operation, respectively.

The former step permits us to drastically mitigate the decorrelation noise artifacts affecting a sequence of time-redundant multi-look small baseline interferograms by properly exploiting their temporal relationships. This is straightforwardly carried out by solving a nonlinear minimization problem that exploits the phase information of conventional multi-look small baseline interferograms only, without requiring the implementation of any pixel selection procedure at the full spatial resolution scale and with no need of *a priori* information on the statistics of the involved complex-valued SAR images.

The latter step is aimed at identifying an optimal network of small baseline interferograms forming a reduced triangulation in the temporal/perpendicular baseline plane, to be efficiently used by the exploited EMCF technique. In particular, the interferogram selection is carried out via the maximization of the (average) equivalent coherence values of the small baseline multi-look interferograms belonging to the reduced triangulation and exploits a SA searching strategy.

The effectiveness of the implemented advanced EMCF-SBAS processing chain has fully been demonstrated by extensively investigating three sets of real SAR data acquired by the ENVISAT sensor over three regions where complex surface deformation scenarios are present.

It is worth remarking that the proposed advanced EMCF-SBAS processing chain can be profitably used to analyze SAR data acquired over regions that represent critical test-sites since the interferograms computed over them are affected by severe decorrelation noise artifacts. The proposed approach can be particularly relevant for the valorization of the large archives of SAR images acquired by the first generation C-band (ERS-1/2, ENVISAT, and RADARSAT-1) SAR sensors. Moreover, the impact of the presented procedure on the SAR data acquired by the new X-band (COSMO-SkyMed and TerraSAR-X) systems is worth for future analyses.

As a final remark, we stress that the proposed advanced EMCF-SBAS processing chain is rather straightforward to implement since it only requires complementing the original EMCF-SBAS chain with two additional preprocessing steps. Nevertheless, to allow an easy upgrade of the existing processing chains, based on the exploitation of small baseline multi-look interferograms, we developed two user-friendly Interactive Data Language routines implementing the proposed interferogram noise-filtering and selection routines; the relevant source codes are freely available at (http://www.irea.cnr.it/en/index.php?option=com_k2&view=item&layout=item&id=505&Itemid=171).

Appendix I

Let $(w_1, w_2, \ldots, w_n)$ be a set of nonrandom (known) weights, and let $(\theta_1, \theta_2, \ldots, \theta_n)$ be a random sample of observations of the circular random variable Θ having a weighted mean direction μ, which is defined, by extending what is presented in [50] and [52], as the direction of the weighted mean resultant $\Pi = \sum_{i=1}^{n} w_i \exp[j\theta_i] / \sum_{i=1}^{n} w_i = \rho \exp[j\mu]$, whose amplitude ρ is the so-called weighted mean resultant length. Note that

$$\cos\mu = \frac{\bar{C}}{\bar{R}}, \quad \sin\mu = \frac{\bar{S}}{\bar{R}}, \quad \rho = \bar{R} \tag{A.I.1}$$

where

$$\bar{C} = \frac{\sum_{i=1}^{n} w_i \cos(\theta_i)}{\sum_{i=1}^{n} w_i}, \quad \bar{S} = \frac{\sum_{i=1}^{n} w_i \sin(\theta_i)}{\sum_{i=1}^{n} w_i}, \quad \bar{R} = \sqrt{\bar{C}^2 + \bar{S}^2}. \tag{A.I.2}$$

From (A.I.2), it is straightforward to show that [50]

$$\frac{\sum_{i=1}^{n} w_i \sin[\theta_i - \mu]}{\sum_{i=1}^{n} w_i} = 0$$
$$\frac{\sum_{i=1}^{n} w_i \cos[\theta_i - \mu]}{\sum_{i=1}^{n} w_i} = \bar{R} = \rho. \tag{A.I.3}$$

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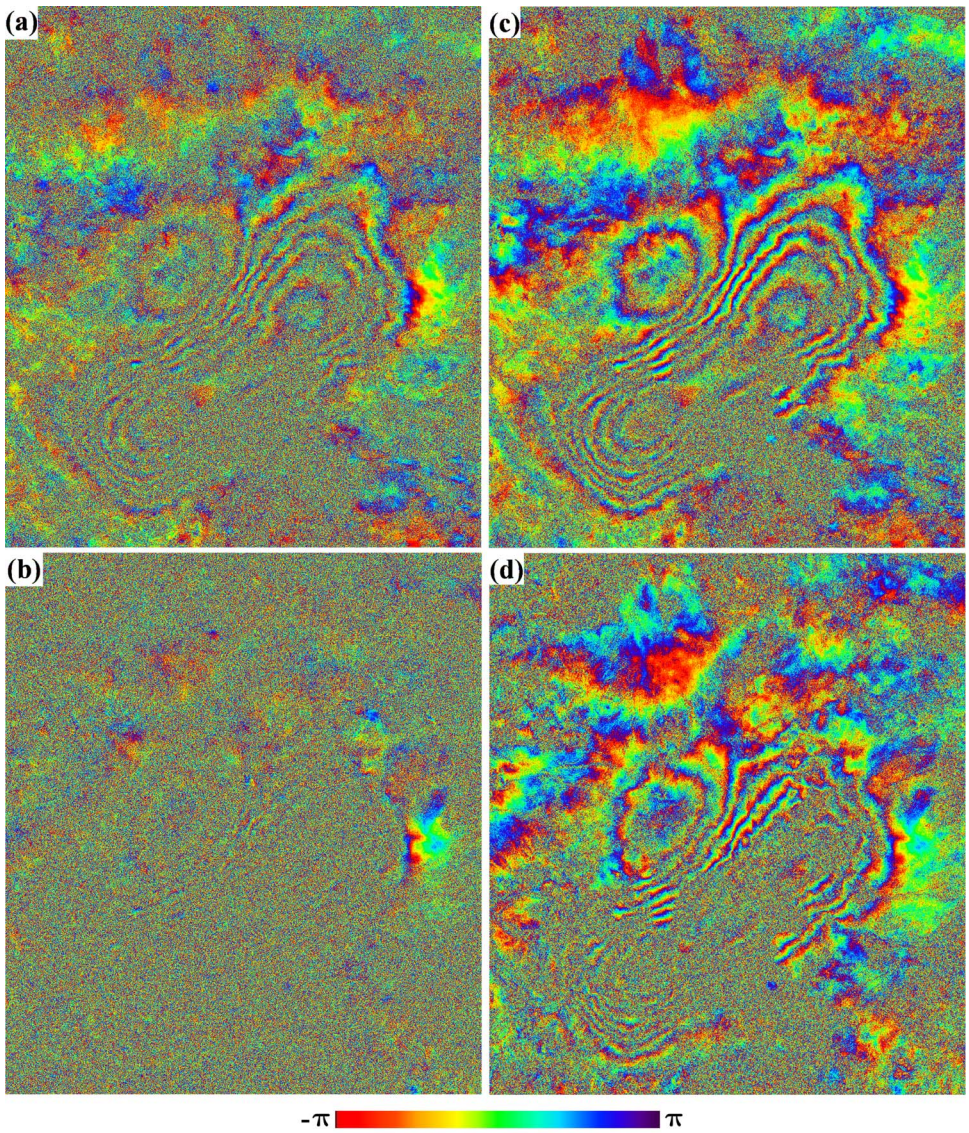


Fig. 18. Comparison between original (left column) and noise-filtered (right column) multi-look interferograms relevant to Yellowstone Caldera. (a) and (b) August 18, 2005, to August 7, 2008, and August 18, 2005, to July 23, 2009, interferograms, characterized by perpendicular baseline values of 40 and 350 m, respectively. (c) and (d) Noise-filtered multi-look interferograms corresponding to the ones in (a) and (b), respectively.

Similarly to [50] and [51], the weighted circular variance ν of the random angle Θ is defined as

$$\nu = 1 - \rho = 1 - \left| \frac{\sum_{i=1}^{n} w_i e^{j\theta_i}}{\sum_{i=1}^{n} w_i} \right| = 1 - \frac{\sum_{i=1}^{n} w_i \cos[\theta_i - \mu]}{\sum_{i=1}^{n} w_i}$$
$$= 1 - \frac{n}{\sum_{i=1}^{n} w_i} E\{\mathbf{w} \cdot \cos[\Theta - \mu]\} \quad \text{(A.I.4)}$$

where $E\{\cdot\}$ is the expectation value operator.

Furthermore, by using $1 - \cos[\chi - \xi]$ to measure the distance between the generic angles χ and ξ, it leads to the use of the following factor:

$$D_w[\alpha] = \frac{\sum_{i=1}^{n} w_i \left[1 - \cos(\theta_i - \alpha)\right]}{\sum_{i=1}^{n} w_i} = 1 - \frac{\sum_{i=1}^{n} w_i \cos(\theta_i - \alpha)}{\sum_{i=1}^{n} w_i} \quad \text{(A.I.5)}$$

as a measure of the dispersion of the weighted circular random signal $\{\mathbf{w} \cdot \exp[j\Theta]\}$ about any direction α. It can be found after some mathematical manipulations, following the lines of [50], that

$$D_w[\alpha] = [1 - \rho] + \rho + \frac{\sum_{i=1}^{n} w_i \cos(\theta_i - \alpha)}{\sum_{i=1}^{n} w_i}$$
$$= \nu + 2\rho \left\{ \sin\left[\frac{\mu - \alpha}{2} \right] \right\}^2. \quad \text{(A.I.6)}$$

Hence, the phase dispersion of the weighted directional data $\{\mathbf{w} \cdot \exp[j\Theta]\}$ is minimal in correspondence to its weighted mean direction, and the minimum value is equal to its weighted circular variance. Notice that $0 \leq \nu \leq 1$. In particular, $\nu = 0$ if and only if the distribution is concentrated around μ; on the

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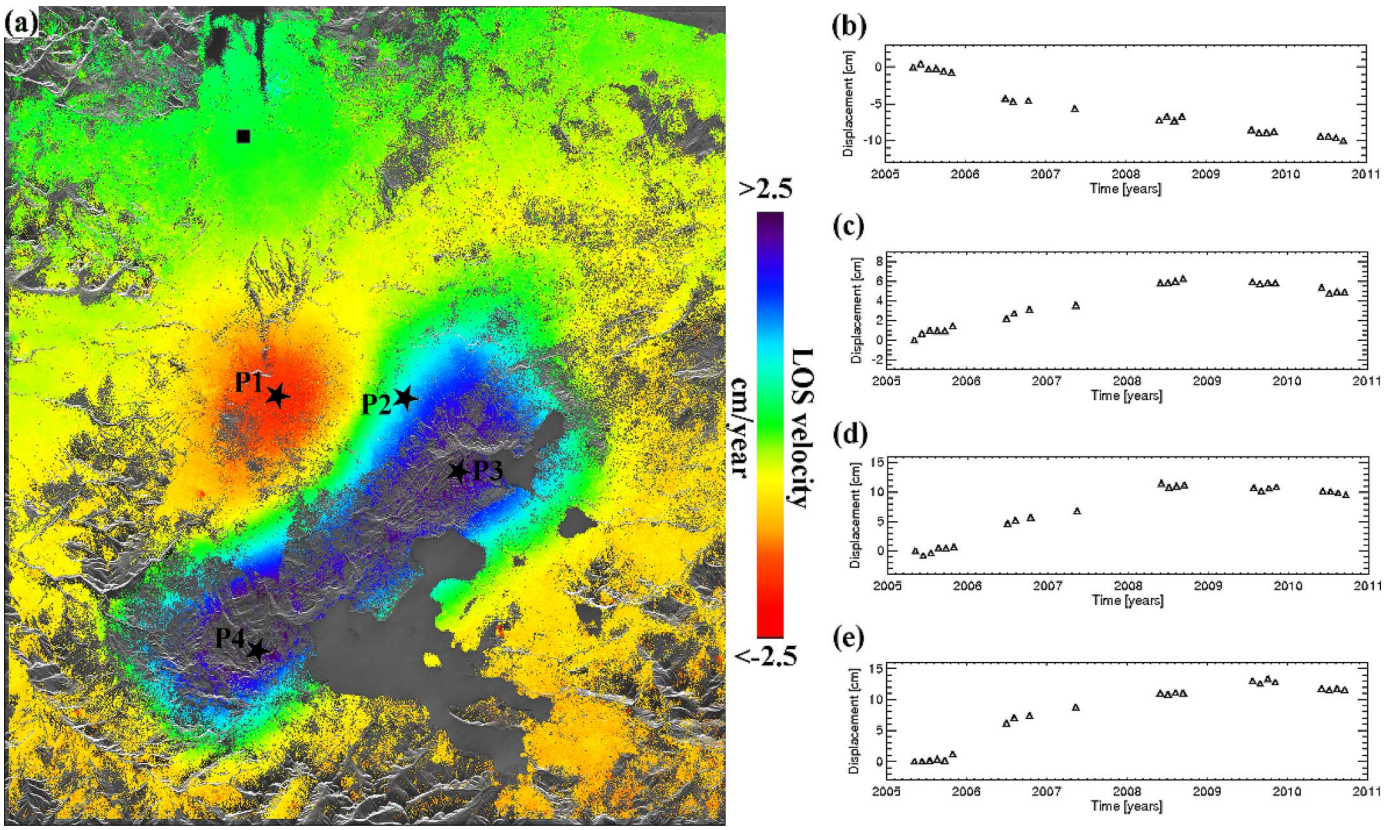


Fig. 19. (a) Mean deformation velocity map (in color) of Yellowstone Caldera, computed in coherent pixels only and superimposed on the SAR amplitude image (gray-scale representation) of the zone, retrieved by applying the advanced EMCF-SBAS processing chain. The black square marks the location of the reference SAR pixel. (b)–(e) Deformation time-series relevant to the four pixels identified via black stars in Fig. 19(a).

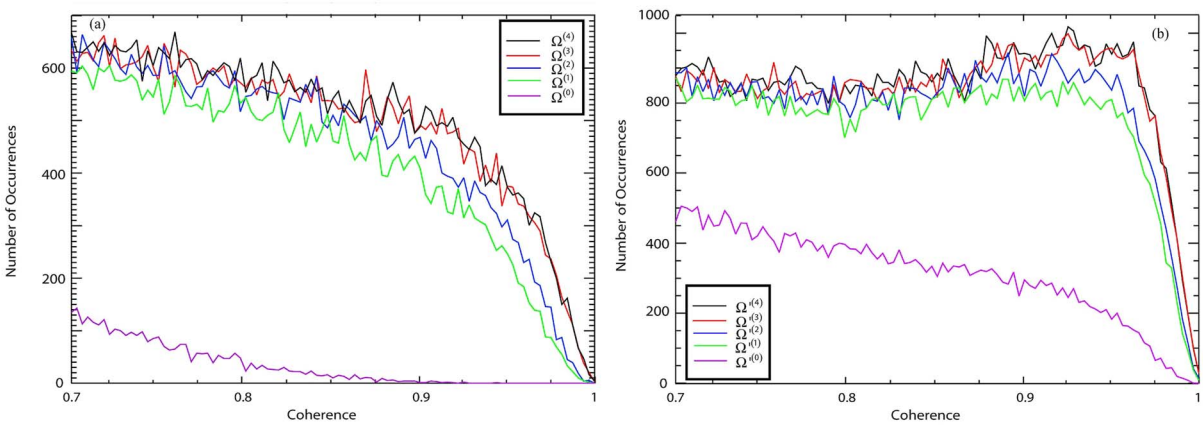


Fig. 20. Impact on the "equivalent" coherence maps computed for the Abruzzi region test-site depending on the number of interferograms exploited by the noise-filtering procedure. (a) Histograms of the equivalent coherence maps computed by considering the whole set of $M^{(\max)}$ interferograms generated by applying the noise-filtering procedure to progressively reduced sets of interferograms depending on the applied baseline constraints. In particular, $\Omega^{(0)}$ is related to the whole set of $M^{(\max)} = 741$ original interferograms (purple line), while $\Omega^{(i)}$ with $i = 1, 2, 3, 4$ are related to the reduced sets of interferograms obtained by imposing: $\Omega^{(1)}$ (green line) no baseline constraints, $M^{(1)} = M^{(\max)} = 741$; $\Omega^{(2)}$ (blue line) $\Delta b_{\perp}^{\max}$800 m, $M^{(2)} = 623$; $\Omega^{(3)}$ (red line) $\Delta b_{\perp}^{\max} = 400$ m, $M^{(3)} = 372$; and $\Omega^{(4)}$ (black line) $\Delta b_{\perp}^{\max} = 400$ m, $\Delta T^{\max} = 2000$ days, $M^{(4)} = 338$. (b) Same as (a) but the equivalent coherence maps ($\Omega'^{(i)}$ $i = 0, 1, 2, 3, 4$) are now computed over the set of $M = M^{(4)} = 338$ small baseline interferograms only.

contrary, if $\nu = 1$, the distribution can be regarded so scattered that there is no concentration around any particular direction. Therefore, these results yield the conclusion that minimizing the (weighted) circular variance allows concentrating as much as possible the distribution of the random signal $\{\mathbf{w} \cdot \exp[j\Theta]\}$ around its mean weighted direction μ.

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APPENDIX II

We discuss in this appendix the similarities and key differences among the noise-filtering technique presented in Section III and those proposed in the phase triangulation [39] and SqueeSAR [40] algorithms.

The main similarity of these methods is relevant to their starting point. Indeed, this is the same as that for [40] and consists in the lack of triangularity among the interferometric phases due to the inconsistencies in the multi-look interferograms, which is expressed through (1). However, we also remark that the aforementioned lack of triangularity information was already addressed and exploited within the InSAR context, well before the publication of [40]. In particular, the lack of triangularity among the multi-look phases was first investigated and exploited in [19, eqs. (9) and (10)] within the temporal PhU strategy adopted to unwrap a sequence of M multi-look interferograms.

Note also that the aforementioned phase inconsistency information was also fully investigated in the 2007 Pepe's Ph.D. Thesis (freely downloadable at the following Internet address http://www.fedoa.unina.it/1572/1/Pepe_Ingegneria_Elettronica_e_delle_Telecomunicazioni.pdf) that has subsequently been published in 2009 [47].

Despite the previously discussed similarity, we remark that there are fundamental differences among our approach and those presented in [39] and [40], for what concerns the exploited statistical frameworks and the mathematical formulation of the minimization procedures applied to retrieve the filtered interferograms.

To fully clarify these issues, let us start our discussion by considering [39, Sec. II]; it describes a SAR data model based on the hypothesis of distributed scattering, under which "the pdf of the data may be regarded as being a zero-mean multivariate circular normal distribution." Under the validity of this hypothesis, a pixel-by-pixel (at the full resolution scale) statistical analysis is then needed to identify on the SAR images the distributed scatterers (DS) with a statistically homogeneous behavior. In particular, a small neighborhood (denoted by Ω) of pixels centered at the same slant range-azimuth location in each single-look SAR image is considered. Under the assumption that all of the pixels falling inside Ω are statistically homogeneous (i.e., "all of the unknown phases as well the coherences are constant inside Ω"), their complete statistical characterization only requires the calculation of the covariance matrix [39, eq. (5)]. Using such a statistical model, an ML procedure is subsequently implemented. This leads to the calculation of the pdf of the data conditioned on the interferometric phases [39, eq. (6)] and, finally, to the ML estimate of the interferometric phases achieved in [39, eq. (12)]. This equation is essentially the same reported in [40, eq. (13)].

Therefore, both the phase Linking [39] and the phase triangulation of the SqueeSAR [40] algorithms require a preliminary identification of the statistically homogeneous pixels (SHPs) on the full-resolution range-azimuth grid. In particular, in [40], the selection strategy of these pixels is based on the application of the Kolmogorov–Smirnov test to carefully select a homogeneous statistical population. Clearly, this requires working at the full resolution spatial scale and implies the analysis of the amplitude values of the complex SAR image pixels.

Let us now consider the statistical framework of our noise-filtering technique. In this case, as said, we exploit conventional multi-look interferograms that are generated by averaging all of the pixels, e.g., N, falling inside a region Ω. Accordingly, at variance with what is assumed in [39] and implemented in [40], we do not carry out any *a priori* selection of the homogeneous pixels that share the same statistical characteristics. As a consequence, the computed multi-look interferograms are characterized by statistics that significantly deviate (see [59] and [60] for an extensive discussion) from those exploited by [39] and [40], which are only valid if the SHPs are preliminarily identified.

Accordingly, our approach and the ones discussed in [39] and [40] are fundamentally different for what concerns the exploited statistical frameworks.

Let us now concentrate on the mathematical formulation of the minimization procedures applied to retrieve the filtered interferograms.

To do this, we highlight that the proposed noise-filtering approach is based on searching for the acquisition phases, namely, $\hat{\Phi}$, which allow minimizing the weighted circular variance, namely, ν, of the residual interferograms, which is the difference between the (multi-look) original $\Delta\Phi$ and reconstructed $\Delta\hat{\Phi}$ interferograms [see (2a) and (2b) that are reported here for the sake of completeness

$$\begin{aligned}\hat{\Phi} &= \arg\min\left[\nu(\hat{\Phi})\right] = \arg\min\left[1-\left|\Pi(\hat{\Phi})\right|\right] \\ &= \arg\max\left[\left|\Pi(\hat{\Phi})\right|\right] \\ \Pi(\hat{\Phi}) &= \sum_{k=1}^{M} c_k \exp[j\cdot r_k] \Big/ \sum_{k=1}^{M} c_k \end{aligned} \tag{A.II.1}$$

where $j=\sqrt{-1}$, $r_k = \langle \Delta\Phi_k - (\hat{\Phi}_{IM_k} - \hat{\Phi}_{IS_k})\rangle_{-\pi,\pi}$, M is the number of small baseline interferograms, and $\{c_k\}_{k=1}^{M}$ are the corresponding coherences.

Conversely, the phase triangulation of the SqueeSAR [40] (as well as the phase linking [39]) algorithm searches for the solution of the following problem:

$$\begin{aligned}\hat{\Phi} &= \arg\max\left[e^{-j\Phi}(\mathbf{A}\circ\hat{\Gamma})e^{j\Phi}\right] \\ &= \arg\max\left\{\mathrm{Re}\left[\sum_{k=1}^{Mtot} a_k c_k \exp(j\cdot r_k)\right]\right\}\end{aligned} \tag{A.II.2}$$

where $\mathbf{A} = |\hat{\Gamma}|^{-1} = |\mathbf{C}|^{-1}$ is the inverse of the coherence matrix, i.e., the modulus of the sample covariance matrix $\hat{\Gamma} = 1/|\Omega|\sum_{P\in\Omega} d(P)\cdot d(P)^H$, with $\mathbf{d}(P) = [d_0(P),\ldots,d_N(P)]^T$ being the complex-valued data vector corresponding to the generic pixel P of the single-look SAR image stack. Note that the symbols ^{H} and $^\circ$ indicate the Hermitian conjugation and the Hadamard (i.e., entrywise)

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product, respectively, Ω is the set of SHP used for the estimation of the covariance matrix, and M_{tot} is the number of all possible interferograms averaged over Ω.

We remark that, although in both approaches the coherence values are exploited, our method uses them as weighting factors, whereas the SqueeSAR method employs the coherence matrix C of all of the interferograms and requires the calculation of its inverse in [40, eq. (14)].

Accordingly, the two optimization schemes are different, and our solution cannot be simply viewed as a merely particularization of [40, eq. (14)] to the case when all of the coherences that are less than a given threshold (corresponding to large baseline interferograms) are set to zero. To clarify this issue, we underline that the phase triangulation algorithm [40, eq. (14)] requires the calculation of matrix A; in this case, even if we put several zeros in matrix C in correspondence to the large baseline interferograms, we would not simply obtain, via the phase triangulation method, the solution that we find through our approach. In addition, the introduction of several zeros in the sample coherence matrix would make it deviate from the "true" coherence matrix, which is the one required to correctly formulate the ML problem [40, eq. (13)] from a statistical point of view.

Note also that the inverse of matrix C, which may be rather critical to compute, is strictly required to implement the phase triangulation method [40], whereas in our approach, this operation is not needed. Indeed, as the SqueeSAR [40] authors correctly said: "even if all of the interferograms and relevant coherences are used, the matrix of sample coherences C is not, in general, positive and definite," thus requiring the introduction of a damping factor to make it invertible. The latter operation is evidently more critical in case several zeros are put into the sample coherence matrix prior to its inversion.

To conclude, in spite of the similarity due to the exploited lack of interferogram triangularity [see (1)], our method is not a mere particularization of [39] and [40] to the small baseline context. Indeed, it is fundamentally different from them for what concerns: 1) the exploited statistical frameworks and 2) the mathematical formulation of the minimization procedure applied to retrieve the filtered interferograms.

APPENDIX III

We present in this appendix a number of experiments showing that the interferogram noise-filtering approach presented in Section III has better performance while applied to sequences of multi-look small baseline interferograms instead of using the whole set of interferograms computed without constraints on the perpendicular and temporal baselines. Starting from the available $N+1=39$ SAR images of the Abruzzi region [see Fig. 3(b)] investigated in Section III, we generated the whole sequence of $M^{(\max)}=(N+1)\cdot N/2=741$ multi-look (original) interferograms, namely, $\Delta\Phi^{(0)}=[\Delta\Phi_1^{(0)},\Delta\Phi_2^{(0)},\ldots,\Delta\Phi_{M^{(\max)}}^{(0)}]^T$. The interferograms $\Delta\Phi^{(0)}$, computed by performing complex multi-look [38] and preliminary noise-filtering [46] operations, were used to retrieve, via the filtering approach discussed in Section III, the corresponding sequence of noise-filtered interferograms, e.g., $\Delta\widehat{\Phi}^{(1)}=[\Delta\widehat{\Phi}_1^{(1)},\Delta\widehat{\Phi}_2^{(1)},\ldots,\Delta\widehat{\Phi}_{M^{(\max)}}^{(1)}]^T$, consisting of $M^{(1)}=M^{(\max)}$ interferograms.

To investigate the quality improvement of the noise-filtered interferograms with respect to the original ones when the entire set of possible interferograms is considered, we selected as a quality index of the achieved results the (equivalent) coherence factor in (6) that is, in this case, computed over the $M^{(\max)}$ interferograms (instead of considering only the ones belonging to a reduced triangulation)

$$\Omega^{(0)}(P)=\frac{\left|\sum_{k=1}^{M^{(\max)}}\exp\left[j\left(\Delta\Phi_k^{(0)}(P)-\Delta\Phi_k^{(0)(LP)}(P)\right)\right]\right|}{M^{(\max)}}\quad\forall\,P\in A_Z\times R_G \tag{A.III.1a}$$

$$\Omega^{(1)}(P)=\frac{\left|\sum_{k=1}^{M^{(\max)}}\exp\left[j\left(\Delta\widehat{\Phi}_k^{(1)}(P)-\Delta\widehat{\Phi}_k^{(1)(LP)}(P)\right)\right]\right|}{M^{(\max)}}\quad\forall\,P\in A_Z\times R_G \tag{A.III.1b}$$

where $\Delta\widehat{\Phi}^{(0)(LP)}$ and $\Delta\widehat{\Phi}^{(1)(LP)}$ are the sequences of the spatial low-pass phase components associated to the original and noise-filtered interferograms, respectively. The histograms of the so-computed $\Omega^{(0)}(\cdot)$ and $\Omega^{(1)}(\cdot)$ equivalent coherence maps are superimposed in Fig. 20(a) (see the purple and green lines, respectively); the coherence improvement is extremely evident.

At this stage, we repeatedly applied the noise-filtering procedure to progressively reduced sets of interferograms, simply selected by imposing constraints on their maximum allowed perpendicular $\Delta b_\perp^{\max}$ and temporal $\Delta T^{\max}$ baseline values. In particular, we performed three tests by considering the three distinctive cases characterized by the following interferometric baseline constraints: 1) $\Delta b_\perp^{\max}=800$ m; 2) $\Delta b_\perp^{\max}=400$ m; and 3) $\Delta b_\perp^{\max}=400$ m, $\Delta T^{\max}=2000$ days, leading to identify a number of interferograms equal to $M^{(2)}=623$, $M^{(3)}=372$, and $M^{(4)}=338$, respectively.[10] For each test, we regenerated the whole set of $M^{(\max)}$ noise-filtered interferograms, namely, $\Delta\widehat{\Phi}^{(2)}$, $\Delta\widehat{\Phi}^{(3)}$, and $\Delta\widehat{\Phi}^{(4)}$, and we computed the corresponding equivalent coherences, namely, $\Omega^{(2)}(\cdot)$, $\Omega^{(3)}(\cdot)$, and $\Omega^{(4)}(\cdot)$, whose histograms are also superimposed in Fig. 20(a) (see the blue, red, and black lines, respectively). By observing Fig. 20(a), it is clear how the progressively reduced number of SAR data pairs does not corrupt the quality of the noise-filtered interferograms used within the noise-filtering procedure, but it is even improved as we discard large baseline interferograms that, hence, do not contribute to the solution with valuable information.

To emphasize the coherence improvement associated to the set of $M=M^{(4)}=338$ small baseline interferograms, we

[10] For the last test, i.e., $M^{(4)}=338$, we used the same baseline constraints and, therefore, the same set of interferograms exploited in the experiments presented in Section III-B.

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identified within the original $\Delta\boldsymbol{\Phi}^{(0)}$ and noise-filtered $\Delta\widehat{\Phi}^{(1)}$, $\Delta\widehat{\Phi}^{(2)}$, $\Delta\widehat{\Phi}^{(3)}$, and $\Delta\widehat{\Phi}^{(4)}$ sequences of interferograms the ones, namely, $\Delta\widehat{\Phi}'^{(1)}$, $\Delta\widehat{\Phi}'^{(2)}$, $\Delta\widehat{\Phi}'^{(3)}$, and $\Delta\widehat{\Phi}'^{(4)}$, corresponding to the 338 small baseline SAR data pairs, and starting from them, we recomputed the corresponding equivalent coherences

$$\Omega'^{(0)}(P) = \frac{\left|\sum_{k=1}^{M} \exp\left[j\left(\Delta\Phi_k'^{(0)}(P) - \Delta\Phi_k'^{(0)(LP)}(P)\right)\right]\right|}{M} \times \forall\, P \in A_Z \times R_G \qquad \text{(III.2.a)}$$

$$\Omega'^{(i)}(P) = \frac{\left|\sum_{k=1}^{M} \exp\left[j\left(\Delta\widehat{\Phi}_k'^{(i)}(P) - \Delta\widehat{\Phi}_k'^{(i)(LP)}(P)\right)\right]\right|}{M} \times \forall\, P \in A_Z \times R_G,\quad i = 1,2,3,4. \qquad \text{(III.2.b)}$$

The histograms of $\Omega'^{(0)}(\cdot)$, $\Omega'^{(1)}(\cdot)$, $\Omega'^{(2)}(\cdot)$, $\Omega'^{(3)}(\cdot)$, and $\Omega'^{(4)}(\cdot)$ are superimposed in Fig. 18(b); also in this case, the obtained improvement is evident.

Accordingly, we may conclude that, to obtain reliable results, it is better to deal with M small baseline interferograms, as described in Section III, instead of using the whole set of interferograms. As a final remark, we observe that more extensive analyses are required to identify the optimal small baseline thresholds: this issue, potentially requiring the processing of several independent SAR data sets, is a matter for future studies.

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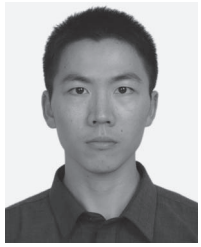
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